

Course Introduction

Data Science & AI

Sabine De Vreese Lieven Smits Bert Van Vreckem
2023–2024

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Contents

Course guide

Introduction

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Course guide

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Learning goals

- Descriptive Statistics
 - Knows some descriptive measures for data (central tendency, dispersion).
 - Can calculate some descriptive measures for data using statistical software (Python).
- Data visualisation
 - Knows different types of plots to represent data visually.
 - Can visualize data using appropriate plots
- Probability
 - Knows the basic rules with regard to calculating with probabilities.
 - Knows the properties of some important probability distributions.

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Learning goals

- Bivariate analysis
 - Can quantify and appropriately test the relationship between two variables.
 - Can construct a simple linear model to show the relationship between two or more variables.
- Time series analysis
 - Can discuss some common models to predict time series and/or detect anomalies.
 - Can indicate the importance of testing the accuracy of a model in a methodologically correct manner.

Course contents

- Introduction, sampling
- Univariate analysis
- Probability, central limit theorem, statistical testing
- Bivariate analysis: qualitative variables
- Bivariate analysis: qualitative vs. quantitative variables
- Bivariate analysis: quantitative variables
- Time series analysis

Learning materials

Published on Chamilo!

- Lecture slides
- Google Colab (<https://colab.research.google.com>)
- Software (optional!):
 - Python (+libraries)
 - Visual Studio Code
 - Github account, Git client
- Lab assignments with example code (Github):
<https://github.com/HoGentTIN/dsai-labs>

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Teaching methods

- 3 hours per week
- classroom instruction (lecture)
- exercises & lab assignments

Recommendations for learning

- Attend classes!
- Take notes
- Use effective learning strategies
- Make an effort

Planning

Wk	Subject
1	Course intro, sampling
2	
3	Univariate analysis
4	
5	Probability, central limit theorem
6	Statistical testing: z-test
7	Statistical testing: Student t -test
—	<i>Easter holiday</i>

Planning

Wk	Subject
8	Bivariate analysis: χ^2 test, Cramér's V
9	Bivariate analysis: qual. vs. quant. variable two-sample t-test, effect size
10	Bivariate analysis: quantitative variables Linear regression
11	Time series analysis
12	Catch-up session (if needed)

Assessment

Written open book exam, with use of computer. Allowed:

- Slides, course material on Chamilo
- Textbook
- Your notes, solutions to exercises
- Python example code
- Software for statistical analysis (Google Colab, Python, VS Code)

Introduction

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Lies, damn lies & statistics

- Data visualisation mistakes and gaffes are common
- Media outlets, politicians, special interest groups, shady people on Facebook, ...misrepresent or misinterpret objective data to “prove” their point.

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Lies, damn lies & statistics



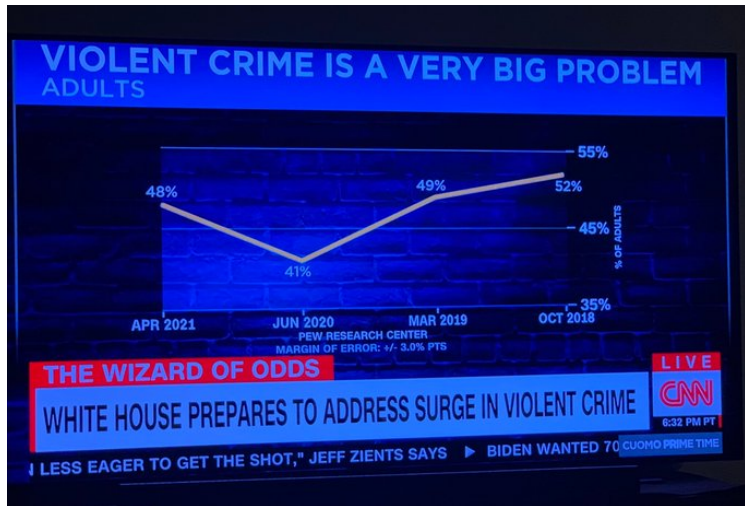
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Lies, damn lies & statistics



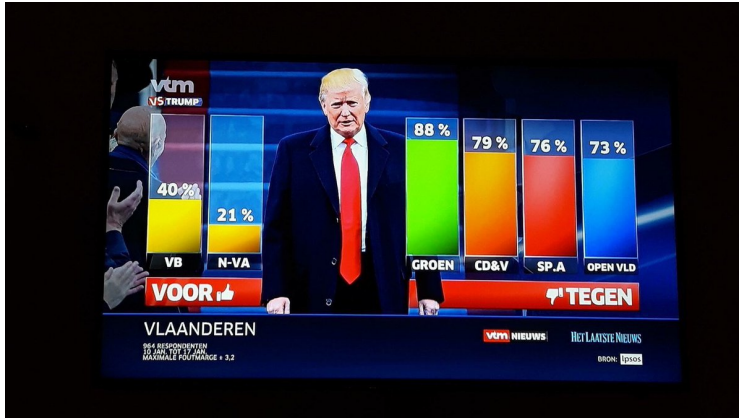
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Lies, damn lies & statistics



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Lies, damn lies & statistics



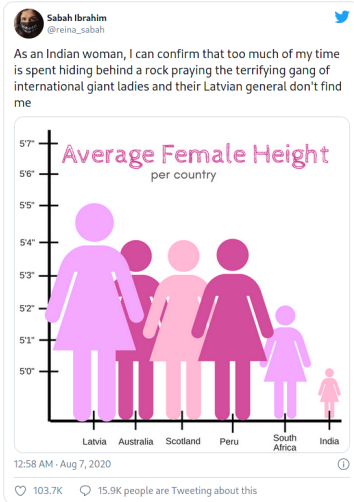
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Lies, damn lies & statistics



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Lies, damn lies & statistics



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Lies, damn lies & statistics

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Het journaal 1 - 21/02/14 Het weer 13.30u 100" Journaal 15u

HOME VIDEOZONE LIVE CENTER

PROGRAMMA'S NIEUWS WAUTERS VS. WAES OOK DAT NOG 60 SECONDEN VK14 OP TV LIVESTREAM MEEST BEKEKEN

VASTGOED
WOONHUIS € 196 000
APPARTEMENT € 207 500

Eigen dak boven hoofd wordt steeds duurder

Andere afleveringen

Het journaal 1 - 21/02/14

Het journaal L - 20/02/14

Het journaal 1 - 20/02/14

Het journaal L - 19/02/14

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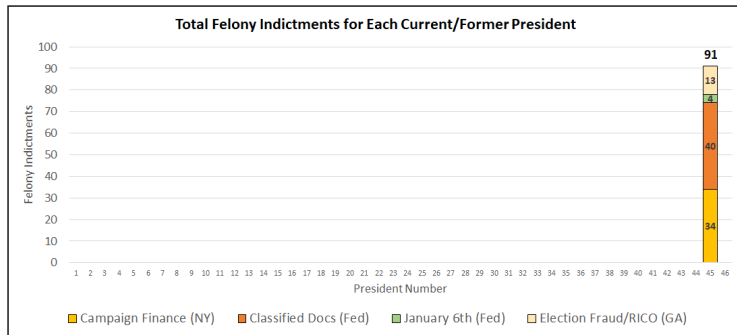
Lies, damn lies & statistics

On average, US presidents have been charged with ≈ 1.978 felony charges.

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Lies, damn lies & statistics

On average, US presidents have been charged with ≈ 1.978 felony charges.



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Why learn data science?

- The amount of data is exploding
- Data drives business decisions
- It's important to analyse and visualise data correctly!
- Tools and data are more accessible than ever

Module 1. Basic concepts and sampling

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2023–2024

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Contents

Basic Concepts in Data Science

Sample Testing

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Learning Goals

- Variables and measurement levels
- Samples
- Basic concepts

Basic Concepts in Data Science

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Variables and Values

Variable General property of an object, allows to distinguish objects

Value Specific property, interpretation for that variable



Variable: gender
Value: male

Variable: hair
Value: bald

Variable: height
Value: 201cm

Variable: weight
Value: 95kg

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Measurement Levels

- = Variable types
- Determine most suitable method for analysis
 - visualization methods
 - central tendency and dispersion
 - examine the relationship between variables

Measurement Levels

Qualitative vs quantitative

Qualitative

Not necessarily numeric
Limited number of values

Quantitative

Number + unit of measurement
Many values, often unique

Quantitative variables often contain the result of a **measurement**

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Measurement Levels

Qualitative scales

Nominal

Categories.

e.g. gender, race, country, shape, ...

Ordinal

Order, rank.

e.g. military rank, level of education, ...

Measurement Levels

Quantitative scales

Interval No fixed zero point $\Rightarrow$ no proportions
e.g. $^{\circ}\text{C}$, $^{\circ}\text{F}$

Ratio Absolute zero point $\Rightarrow$ proportions
e.g. distance (m), energy (J), weight (kg) ...

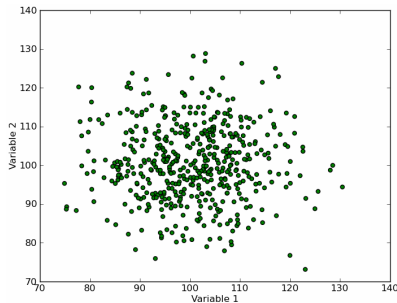
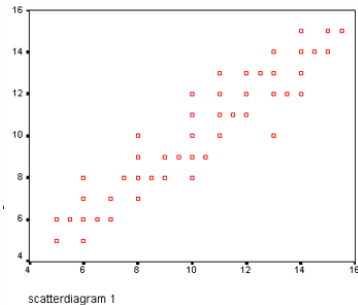
Proportions:

- 20 m is $1/3$ th or $\sim 33\%$ longer than 15 m
- 20 $^{\circ}\text{C}$ is **NOT** $1/3$ th warmer than 15 $^{\circ}\text{C}$ (convert to $^{\circ}\text{F}$)

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Relations between variables

Variables are related if their values change **systematically**.



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Relations between variables: example

Is there a relationship between type of cola and taste appreciation?

	Pepsi	Coca Cola	Total
Like	56	24	80
Dislike	14	6	20
Total	70	30	100



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Relations between variables: example

Is there a relationship between type of cola and taste appreciation?

	Pepsi	Coca Cola	Total
Like	56	24	80
Dislike	14	6	20
Total	70	30	100



Marginal totals

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Causal Relationships

Researchers are often looking for **causal relationships**, e.g.

- Frustration leads to aggression
- Alcohol leads to decreased alertness
- ...

Cause Independent variable

Consequence Dependent variable

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Causal Relationships

Fake correlations or “Spurious correlations”

Warning!

A relationship between variables does not necessarily indicate a *causal* relation!

Examples:

- Violent video games lead to violent behaviour
- Vaccines can cause autism
- Relationship between drinking cola light and obesitas
- ...

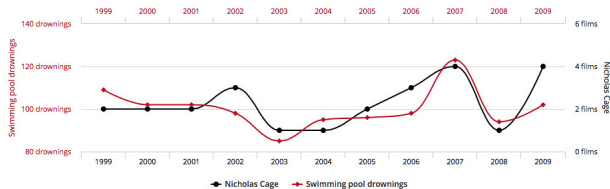


Number of people who drowned by falling into a pool

correlates with

Films Nicolas Cage appeared in

Correlation: 66.6% ($r=0.666004$)



Data sources: Centers for Disease Control and Prevention and Internet Movie Database

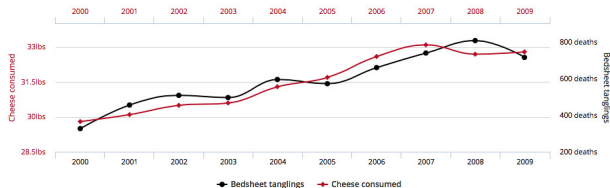
tylervigen.com

Per capita cheese consumption

correlates with

Number of people who died by becoming tangled in their bedsheets

Correlation: 94.71% ($r=0.947091$)



Data sources: U.S. Department of Agriculture and Centers for Disease Control and Prevention

tylervigen.com

Sample Testing

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USA Today has come out with a new survey. Apparently, three out of every four people make up 75% of the population

—David Letterman

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Suppose you want to analyze a group of friends

Questions you can ask:

- How tall are my friends?
- What are their weights?
- How safe is their living environment?
- Do they have family?
- ...

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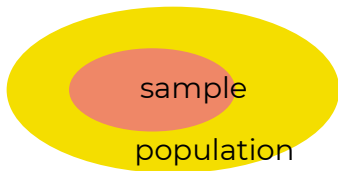
Population



Sample and Population

Population the collection of all objects/people/...that you want to investigate

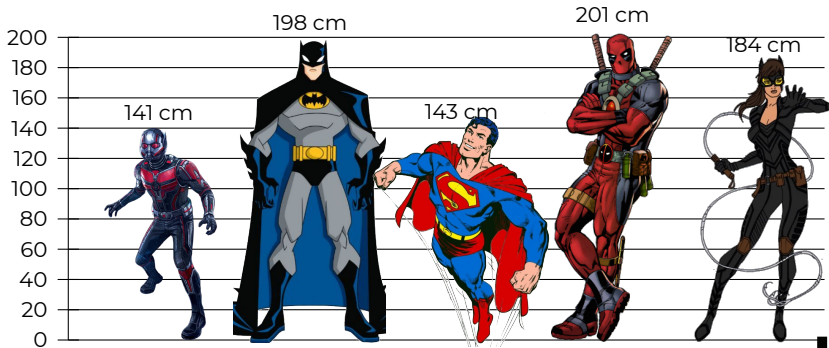
Sample a *subset* of the population from which measurements will be taken



Under certain circumstances, the results for a sample are representative for the population.

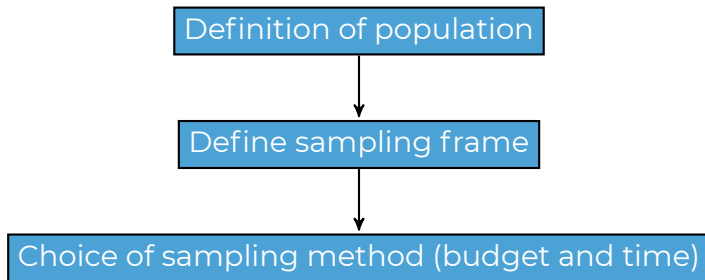
Sample and Population

A sample is easier to analyze than the entire population



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Sampling Method



How to pick elements for a sample?

Random sample : every element from the population has an equal chance of being included in the sample.

Non-random sample : the elements for the sample are *not* randomly selected. Objects that can be collected *easily* are more likely to be included (convenience sampling).



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Stratified to variables

Gender	Age				Total
	≤ 18	]18,25]	]25,40]	> 40	
Woman	500	1500	1000	250	3250
Man	400	1200	800	160	2560
Total	900	2700	1800	410	5810

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Stratified to variables

Gender	Age				Total
	≤ 18	]18,25]	]25,40]	> 40	
Woman	500	1500	1000	250	3250
Man	400	1200	800	160	2560
Total	900	2700	1800	410	5810

Gender	Age				Total
	≤ 18	]18,25]	]25,40]	> 40	
Woman	50	150	100	25	325
Man	40	120	80	16	256
Total	90	270	180	41	581

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Possible Errors

Measurements in a sample will typically deviate from the value in the entire population $\Rightarrow$ Errors!

- Accidental $\leftrightarrow$ Systematic
- Sampling error $\leftrightarrow$ Non-sampling error

Sampling Errors

- Accidental sampling errors
 - Pure coincidence

Sampling Errors

- Accidental sampling errors
 - Pure coincidence
- Systematic sampling errors
 - Online survey: people without internet are excluded
 - Street survey: only who is currently walking there
 - Voluntary survey: only interested parties participate

Non-sampling Errors

- Accidental non-sampling errors
 - Incorrectly ticked answers

Non-sampling Errors

- Accidental non-sampling errors
 - Incorrectly ticked answers
- Systematic non-sampling errors
 - Poor or non-calibrated measuring equipment
 - Value can be influenced by the fact that you measure
 - Respondents lie (number of cigarettes a day)

Module 2. Univariate statistics

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2023–2024

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Contents

Central Tendency and Dispersion

Measure of Central Tendency

Measures of Dispersion

Summary

Data visualisation

Simple Graphs

Interpretation of Charts

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Learning Goals

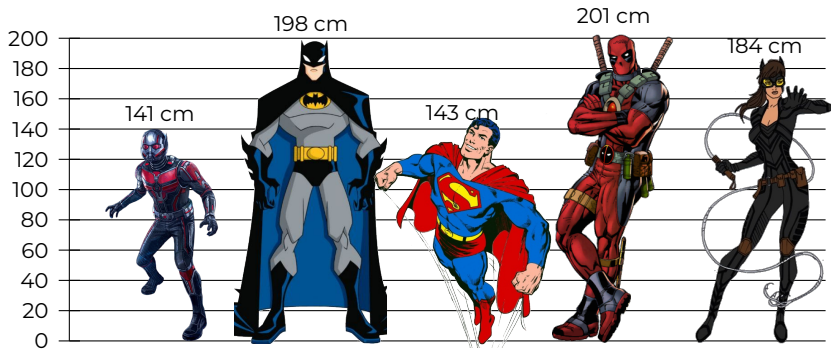
- Descriptive statistics
- Central tendency and dispersion for each measurement level
- Know formulas, being able to calculate
- Suitable visualization techniques for each measurement level

Central Tendency and Dispersion

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How tall are my friends?

Remember our superheroes:



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Measure of Central Tendency

What value is representative of the entire group?

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Mean or Average

Arithmetic mean

The **arithmetic mean** (notation: $\bar{x}$) is the sum of all values divided by the number of values

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

x_1	x_2	x_3	x_4	x_5
141	198	143	201	184



Mean or Average

- Q1** What happens if Ant-Man shrinks to a size of 10 cm?
- Q2** The arithmetic mean of 15 numbers is 12. What number should be added to get a mean of 13?



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Median

Median

To find the **median**, sort all values and pick the middle number

- Odd number of values: no problem
- Even number of values: average of the middle two

x_1	x_2	x_3	x_4	x_5
141	198	143	201	184



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Median

- Q1** What happens if Ant-Man shrinks to a size of 10 cm?
- Q2** What is the median of the number of people saved by Batman during the last eight years?

4	7	11	16	20	22	25	26
---	---	----	----	----	----	----	----



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Mode

Mode

The **mode** is the value that appears most often in a dataset.

Number of people saved by Superman during the last 15 years:

3	7	5	13	20	23	39	23	40	23	14	12	56	23	29
---	---	---	----	----	----	----	----	----	----	----	----	----	----	----



Number of people saved by Batman during the last 8 years:

4	7	11	16	20	22	25	26
---	---	----	----	----	----	----	----



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Measures of Dispersion

How large are the differences within the group?

Range

Range

The **range** of a dataset is the absolute value of the difference between the highest and the lowest value.

x_1	x_2	x_3	x_4	x_5
141	198	143	201	184



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Quartiles

Quartiles

The **quartiles** of a sorted set of numbers are the three values that divide the set into 4 equally large subsets. Notation: Q_1, Q_2, Q_3

Number of people saved by Superman during the last 15 years:

3	7	5	13	20	23	39	23	40	23	14	12	56	23	29
---	---	---	----	----	----	----	----	----	----	----	----	----	----	----



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Calculating Quartiles

- Different software programs have slightly different ways of calculating quartiles.
- The following method is easy to perform by hand. Start by sorting the values.
 - When n is odd.
 - The median (Q_2) is the middle value (as before).
 - Leave out the median. Q_1 is the median of the first half, Q_3 is the median of the second half.
 - When n is even.
 - The median (Q_2) is the average of the two middle values.
 - Q_1 is the median of the first half, Q_3 is the median of the second half.

Interquartile Range (IQR)

Interquartile Range

The **interquartile range** is the difference between the third and first quartile $|Q_3 - Q_1|$.

Variance and Standard Deviation

Variance

The **variance** (s^2 or σ^2) is the mean squared difference between the values of a data set and the arithmetic mean.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Standard deviation

The **standard deviation** (s or σ) is the square root of the variance

x_1	x_2	x_3	x_4	x_5
141	198	143	201	184



Properties of the Standard Deviation

- Can the standard deviation be negative?

Properties of the Standard Deviation

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- What is the smallest possible value? What does this imply?

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- What is the unit of the standard deviation (in relation to the unit of the variable)?

Properties of the Standard Deviation

- Can the standard deviation be negative?
- What is the smallest possible value? What does this imply?
- What effect do outliers have on the standard deviation?
- What is the unit of the standard deviation (in relation to the unit of the variable)?
- How do you interpret the standard deviation combined with the average?

Properties of the Standard Deviation

Why $n - 1$ in the denominator and not n ?

You can prove the reason for the change mathematically, but we will investigate it empirically

See Python example code in `demo-analysis-1-var.ipynb`



Het journaal 1 - 21/02/14



Het weer 13.30u



100'' Journaal 15u

HOME

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PROGRAMMA'S ▼

NIUWS ▼

WUTERS VS. WAES

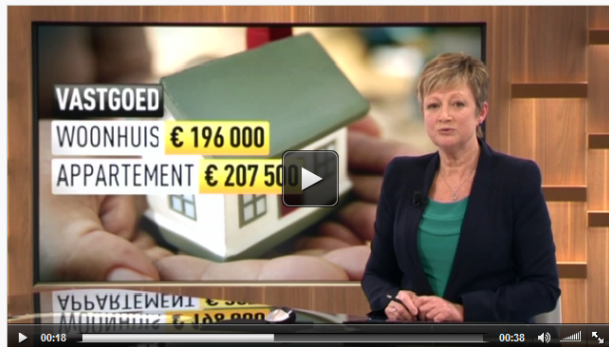
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60 SECONDEN

VK14 OP TV

LIVESTREAM

MEEST BEKEKEN



Eigen dak boven hoofd wordt steeds duurder

Andere afleveringen



Het journaal 1 - 21/02/14



Het journaal L - 20/02/14



Het journaal 1 - 20/02/14



Het journaal L - 19/02/14

This news item reports on high prices for houses and flats. Do the numbers give a good idea of the situation?

Remember!

Providing only a center value is never sufficient!

- What is the dispersion?
- How is the data distributed? Normal distribution?
- Is the group sufficiently homogeneous?

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Central Tendency and Dispersion: Summary

Measurement Level	Center	Spread Distribution
Qualitative	Mode	—
Quantitative	Average/Mean Median	Variance, Standard Deviation Range, Interquartile Range

Summary of Symbols

	Population	Sample
number of elements	N	n
average or mean	μ	$\bar{x}$
variance	$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$	$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$
standard deviation	σ	s

Data visualisation

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Chart type overview

Measurement level	Chart type
Qualitative	Bar chart
Quantitative	Boxplot Histogram Density plot

See Python-example code in `demo-analysis-1-var.ipynb`

Pie Chart

Attention!

Avoid using a pie chart!

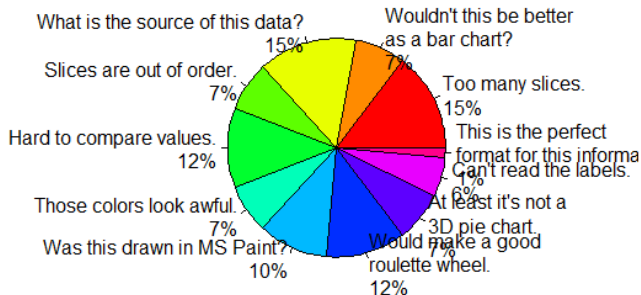
Disadvantages:

- Comparing angles is harder than comparing length
- Unusable for data with many categories

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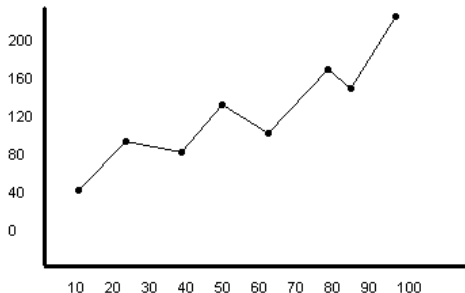
Pie Chart

What people are saying about your pie chart



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Interpretation of Charts



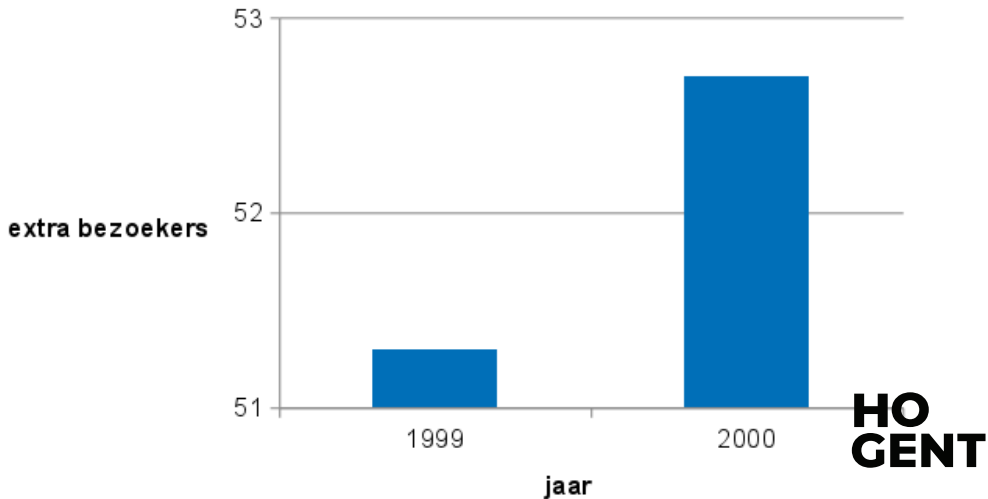
Tips:

- Label the axes
- Add a clear title
- Name the unit (and, if necessary, order of magnitude)
- Add a label that clarifies the chart

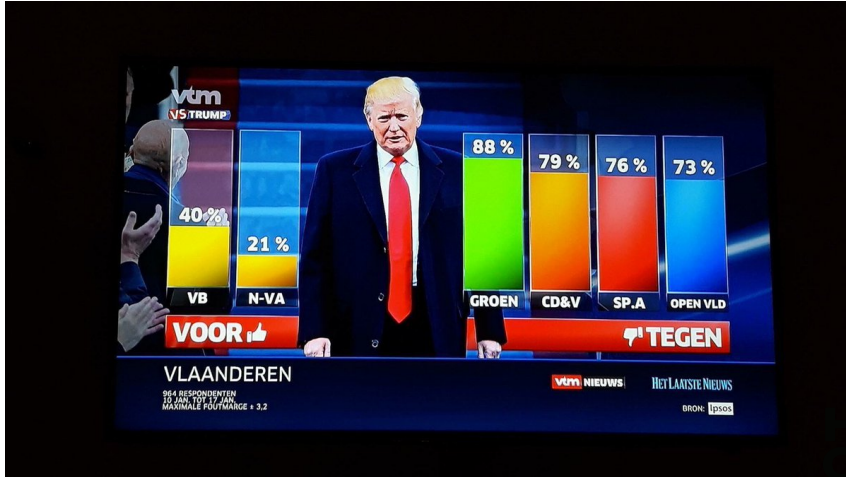
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Data distortion

= misrepresenting data so that invalid conclusions are drawn



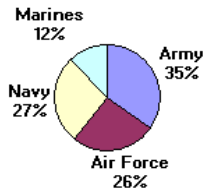
Data distortion



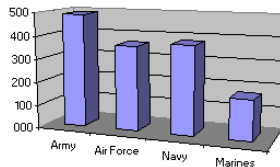
Data distraction

- Avoid bells and whistles
- Minimize “ink to data” ratio

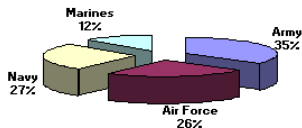
Active Duty Personnel, 1998



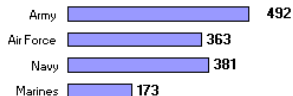
Active Duty Personnel, 1998
(millions)



Active Duty Personnel, 1998



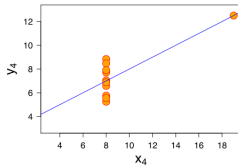
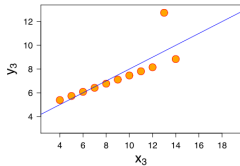
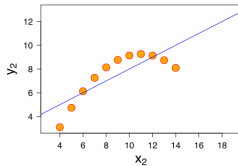
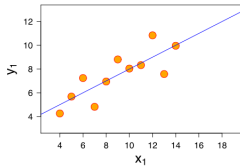
Active Duty Personnel, 1998
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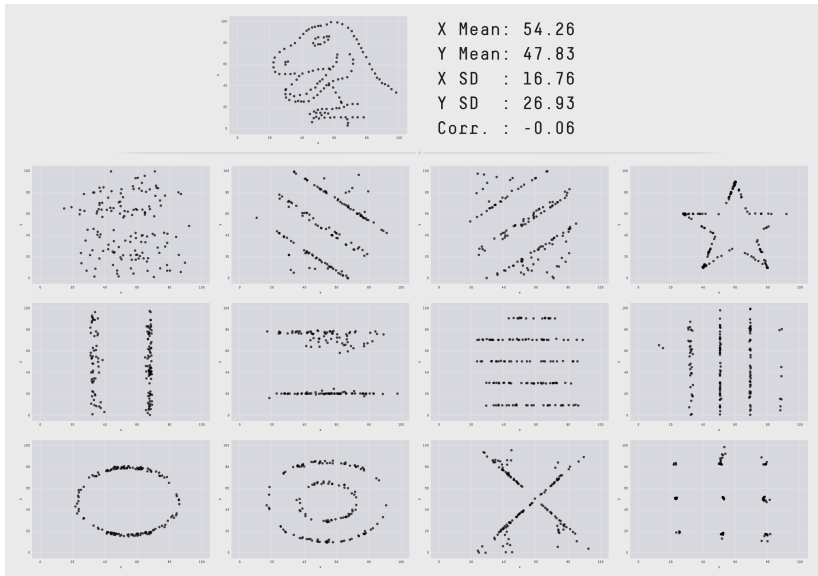
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The importance of data visualization

Anscombe's Quartet are four completely different datasets with the same measurements of central tendency and dispersion.



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“The Datasaurus Dozen” (Source:
<https://www.autodeskresearch.com/publications/samestats>)

Module 3a. Probability, The Central Limit Theorem, Confidence Intervals

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Contents

Probability

- Discrete random variables

- Expectation and variance of a random variable

- Continuous Random Variables

- The (Standard) Normal Distribution

- Probabilities in the Normal Distribution

- Other Continuous Distributions

From Sample to Population

- The Central Limit Theorem

- Confidence Intervals

- Confidence Interval for a Large Sample

- Confidence Interval for a Small Sample

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Learning Goals

- Probability distribution of a sample
- The normal and Student- t distribution
- Formulate the central limit theorem and explain its importance for statistical analysis
- Calculate confidence intervals

Probability

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Probability

Informally, probabilities represent beliefs of how likely it is that a certain event will happen when performing a certain experiment.

In the frequentist view, **probability** represents the **relative frequency** of the occurrence of the event at hand (when performing a large number of independent experiments).

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Probability: (Simple) Examples

A fair six-sided die is thrown once.



- What is the probability of getting a “1”?
- What is the probability of throwing an even number?
- What is the probability of getting a number in the set $\{1, 2, 3, 4, 5, 6\}$?

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Probability: (Simple) Examples

A fair six-sided die is thrown once.



- What is the probability of getting a “1”?
- What is the probability of throwing an even number?
- What is the probability of getting a number in the set $\{1, 2, 3, 4, 5, 6\}$?

Answers: $1/6$, $1/2$ and 1

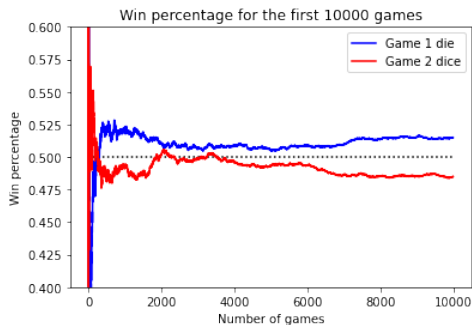
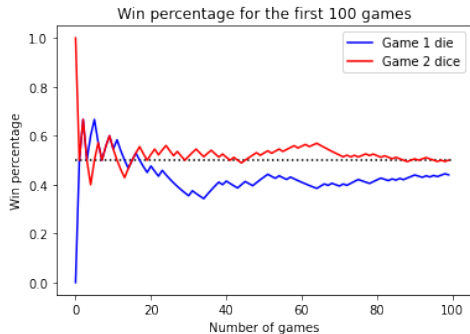
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Probability: Example

Two games of chance:

1. Bet on: at least one six when throwing a fair die 4 consecutive times.
 - o You will *win* in the long run when playing this game.
2. Bet on: at least one “double six” when throwing two fair dice 24 times.
 - o You will *lose* in the long run when playing this game, even though in some sense the two bets seem equivalent. (Probability divided by six but number of throws multiplied by six.)

Probability: Example



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Probability: Observations

- Probabilities are numbers assigned to sets.
- These sets are part of some all-encompassing set, the “universe”, typically denoted Ω .
- The numbers (probabilities) assigned to the sets have to satisfy three simple rules (see later) in order to correspond to our intuition about how probabilities should behave.

Example

For the first game: at least one six when throwing a fair die 4 consecutive times.

- What is the universe?

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- What is the universe? Answer:

$$\Omega = \{(1, 1, 1, 1), (1, 1, 1, 2), (1, 1, 1, 3), \dots, (6, 6, 6, 4), (6, 6, 6, 5), (6, 6, 6, 6)\}$$

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Example

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- What set corresponds to the event: “you win” or “at least one 6”?
Answer

$$A = \{(6, 1, 1, 1), (6, 1, 1, 2), \dots, (5, 5, 5, 6), (6, 6, 1, 1), \dots, (6, 6, 5, 5), (1, 6, 6, 1), \dots, (6, 6, 6, 6)\}$$

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Axioms (Rules) of Probability

Axioms of Probability

1. Probabilities are non-negative: $P(A) \geq 0$ for each A .
2. The universe has probability 1: $P(\Omega) = 1$
3. *Sum rule*: When A and B are *disjoint* events (i.e. $A \cap B = \emptyset$) then it holds that

$$P(A \cup B) = P(A) + P(B)$$

Properties of Probabilities

From the three axioms of probability, we can derive *all* properties of probabilities. Some important ones are listed below:

- **Complement rule:** for each A it holds that

$$P(\bar{A}) = 1 - P(A)$$

where $\bar{A}$ represents the event “ A does not occur”.

- The impossible event has probability zero: $P(\emptyset) = 0$.
- The **general sum rule:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Examples

1. A coin is tampered with such that $P(\text{head}) = 0.6$. What is $P(\text{tail})$?
2. From the 250 students enrolled in this course, 200 own a PlayStation game console, while 100 own a Nintendo Switch. There are 75 students who own both game consoles. What is the probability, when you pick a random student, that
 - o this student does not own a PlayStation
 - o this student owns a PlayStation or a Nintendo Switch (or both)
 - o this student owns neither a PlayStation nor a Nintendo Switch.

Examples

1. A coin is tampered with such that $P(\text{head}) = 0.6$. What is $P(\text{tail})$?
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Answers: $1 - 200/250 = 1/5$, $(200 + 100 - 75)/250 = 225/250 = 9/10$,
 $1 - 9/10 = 1/10$

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Independent Events: Example

When events are **independent** it means that the occurrence of one event (or knowing that the event occurred) **does not change** the probability of the other event occurring.

As an example: draw a card from a standard deck of cards and consider the following events:

A = a ♣ was drawn

and

B = an ace was drawn

When I tell you that B occurred, this will not change your opinion about the probability of A occurring, this will still be $13/52 = 1/4$.

(Note: this also works the other way around!)

Independent Events: Definition

Independent events

Mathematically, two events A and B are independent when

$$P(A \cap B) = P(A)P(B)$$

Dependent Events: Example

Consider the events from before where students own either a PlayStation, a Nintendo Switch or both. Let A be the event that a student owns a PlayStation, while B represents the fact that a student owns a Nintendo Switch. From the numbers above we see that

$$P(A) = \frac{200}{250} = 0.8, \quad \text{and} \quad P(B) = \frac{100}{250} = 0.4$$

when we multiply these two probabilities we get

$$P(A)P(B) = 0.8 \times 0.4 = 0.32$$

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However, the probability that a student owns both a PlayStation and a Nintendo Switch, i.e. $A \cap B$ is given by

$$P(A \cap B) = \frac{75}{250} = 0.3 \neq 0.32$$

Hence,

$$P(A \cap B) \neq P(A)P(B)$$

and the two events are **dependent**.

Dependent Events: Example

In practical terms this means that e.g. knowing that a student owns a PlayStation gives you additional information about his or her ownership of a Nintendo Switch.

Before you know that the student owns a PlayStation the probability that this student owns a Nintendo Switch is

$$P(B) = \frac{100}{250} = 0.4$$

However, once you know that that the student owns a PlayStation you know that this student belongs to the group of 200 PlayStation owners, of which 75 also own a Nintendo Switch. With fancy notation we write this as

$$P(B | A) = \frac{75}{200} = 0.375$$

so the probability now is slightly lower.

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Dependent Events: Example

On the other hand, when you know that a student owns a Nintendo Switch, what is the probability of this student owning a PlayStation?

Dependent Events: Example

On the other hand, when you know that a student owns a Nintendo Switch, what is the probability of this student owning a PlayStation?

Answer:

$$P(A | B) = \frac{75}{100} = 0.75$$

Discrete Random Variable

Example

A player draws a card from a standard deck of cards. She gets 1 € when drawing a jack, 2 € when drawing a queen and 3 € when drawing a king. In all other cases she receives nothing.

We can formalise this by defining a mapping X from Ω to $\mathbb{R}$:

$$X: \Omega \rightarrow \mathbb{R} : \omega \mapsto X(\omega) = \begin{cases} 1 & \text{when drawing a jack} \\ 2 & \text{when drawing a queen} \\ 3 & \text{when drawing a king} \\ 0 & \text{in all other cases.} \end{cases}$$

We have done nothing else but associate a *number* with different events.

Probability Mass Function

Using the rules of probability we can assign probabilities to the different numerical outcomes

$$f_X(3) = P(X = 3) = P(\text{king}) = \frac{4}{52} = \frac{1}{13},$$

$$f_X(2) = P(X = 2) = P(\text{queen}) = \frac{4}{52} = \frac{1}{13},$$

$$f_X(1) = P(X = 1) = P(\text{jack}) = \frac{4}{52} = \frac{1}{13},$$

$$f_X(0) = P(X = 0) = P(\text{any other card}) = \frac{40}{52} = \frac{10}{13}.$$

Note:

- It holds that $f_X(0) + f_X(1) + f_X(2) + f_X(3) = 1$.
- And also: $0 \leq f_X(x_i) \leq 1$.

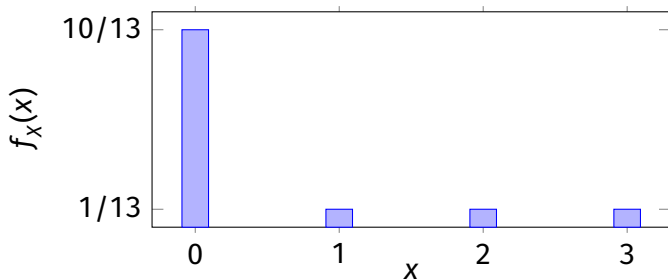
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Probability Mass Function

To make things more explicit we can put the probability mass function in a table:

x	0	1	2	3
$f_X(x) = P(X = x)$	$\frac{10}{13}$	$\frac{1}{13}$	$\frac{1}{13}$	$\frac{1}{13}$

We can also represent this probability mass function in a graph:



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Probability Distribution: 1 die

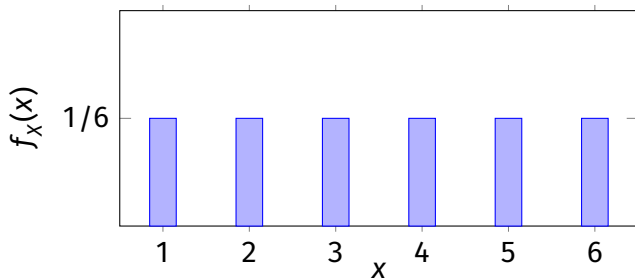
What is the probability of each outcome when rolling a single die?

x	1	2	3	4	5	6

Probability Distribution: 1 die

What is the probability of each outcome when rolling a single die?

x	1	2	3	4	5	6
$f_X(x) = P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$



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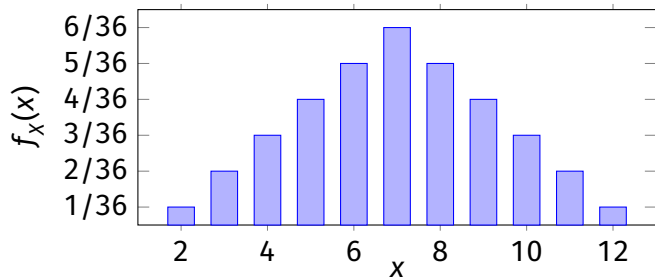
Probability Distribution: 2 dice

x	2	3	4	5	6	7	8	9	10	11	12

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Probability Distribution: 2 dice

x	2	3	4	5	6	7	8	9	10	11	12
$f_X(x) = P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$



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Using Probability Mass Func.

Once you have a probability mass function, you can compute other probabilities.

With the card game, what is the probability that you gain at most 2 EUR?

Using Probability Mass Func.

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With the card game, what is the probability that you gain at most 2 EUR?
Answer:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{10}{13} + \frac{1}{13} + \frac{1}{13} = \frac{12}{13}$$

We could also have used the complement rule in this case:

$$P(X \leq 2) = 1 - P(X > 2) = 1 - P(X = 3) = 1 - \frac{1}{13} = \frac{12}{13}.$$

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Using Probability Mass Func.

When throwing two dice, what is the probability that the sum of the results is between 4 and 7, boundaries included

Using Probability Mass Func.

When throwing two dice, what is the probability that the sum of the results is between 4 and 7, boundaries included

Answer:

$$P(4 \leq X \leq 7) = P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) = \frac{18}{36} = \frac{1}{2}$$

We could also compute this using left-tail probabilities only:

$$P(4 \leq X \leq 7) = P(X \leq 7) - P(X < 4) = P(X \leq 7) - P(X \leq 3) = \frac{21}{36} - \frac{3}{36} = \frac{1}{2}$$

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Expectation of a R.V.

Consider the random variable associated with playing the card game with the following probability distribution function

$$\begin{aligned}f_X(3) &= P(X = 3) = \frac{1}{13}, & f_X(2) &= P(X = 2) = \frac{1}{13}, \\f_X(1) &= P(X = 1) = \frac{1}{13}, & f_X(0) &= P(X = 0) = \frac{10}{13}.\end{aligned}$$

If somebody played this game a very large number of times, what would their average earnings be? I.e. what is the **expected value** of this random variable?

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Expectation of a R.V.

Expectation of a random variable

The expectation of a random variable is denoted by μ_X or $E(X)$ and is given by

$$\mu_X = \sum_i x_i P(X = x_i) = \sum_i x_i f_X(x_i).$$

Note the similarity between this formula and the way you calculate the sample mean when a frequency table is given.

We often write μ instead of μ_X .

Expectation of a R.V.

The expected value of the random variable associated with the card game is given by:

$$\begin{aligned}\mu_X &= \sum_i x_i f_X(x_i) \\ &= 0 \cdot f_X(0) + 1 \cdot f_X(1) + 2 \cdot f_X(2) + 3 \cdot f_X(3) \\ &= 0 \cdot \frac{10}{13} + 1 \cdot \frac{1}{13} + 2 \cdot \frac{1}{13} + 3 \cdot \frac{1}{13} \\ &= \frac{6}{13}.\end{aligned}$$

Variance of a R.V.

The variance of a R.V. is a measure of dispersion that resembles the sample variance very closely.

Variance of a random variable

The variance of a random variable is defined by:

$$\sigma_X^2 = \sum_i (x_i - \mu_X)^2 P(X = x_i) = \sum_i (x_i - \mu_X)^2 f_X(x_i).$$

Note: the **standard deviation** is the positive square root of the variance.

$$\sigma_X = \sqrt{\sigma_X^2}$$

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Variance: Intuition

Consider two random variables X and Y .

The random variable X represents the number of received emails per day for a first office worker. This person receives either 48 or 52 emails per day, each with a 50 percent probability. The random variable Y represents the number of received emails of a second office worker. This second person receives no emails at all with 50 percent probability and with 50 percent chance receives 100 emails.

- Determine the probability mass function of X and of Y .
- Compute the expected value of X and Y . What do you notice?
- Who is experiencing a larger variability w.r.t. the number of emails received?
- Compute the variance of X and Y .

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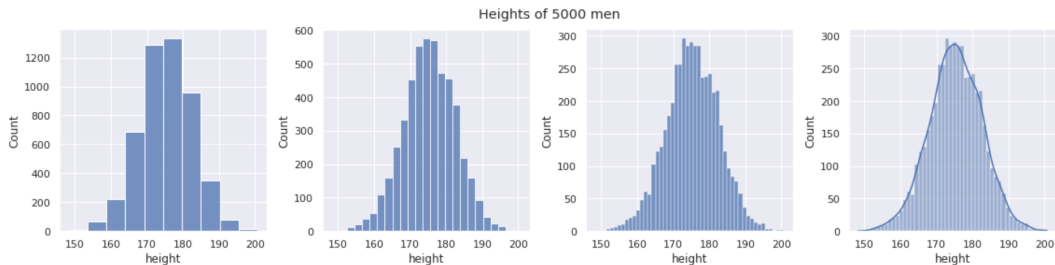
Continuous Random Variable

- A continuous random variable takes on an uncountably infinite number of possible values.
- In this case it doesn't make a lot of sense to consider the probability that X equals some number a **exactly**, because this probability is always zero.
- What does make sense is to consider the probability that X takes on a value in some interval $[a, b]$.
- This probability can be found by *integrating* (i.e. "summing up") the **probability density function** of the random variable.

Continuous Random Variable

Measurements of length, time, ... are examples of continuous random variables.

The leftmost histogram illustrates that most of the 5000 men were between 175cm and 180cm tall.

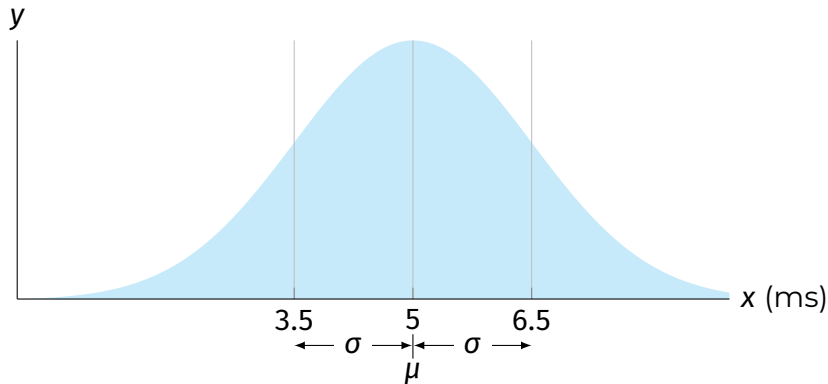


Continuous Random Variable

- From left to right the number of intervals is increased, but the total area of all the bars always equals 1.
- If the number of intervals is increased even more, the intervals would eventually get so small that we no longer have a histogram, but rather a curve (by connecting the 'dots' at the tops of the very tiny rectangles).
- The heights of people often follow an approximate **Normal Distribution**. The Normal Distribution is a type of **continuous probability distribution**.
- There are also other continuous probability distributions, e.g. the Exponential Distribution and the continuous Uniform Distribution.

Continuous Probability Distr.

The reaction speed x of Superman (in ms) can also be represented as a normal distribution:



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Expectation and Variance

Also for continuous R.V.'s there are formulas for expectation and variance.
The expectation is given by

$$\mu_X = \int_{-\infty}^{+\infty} x f_X(x) dx,$$

while the variance σ_X^2 is given by

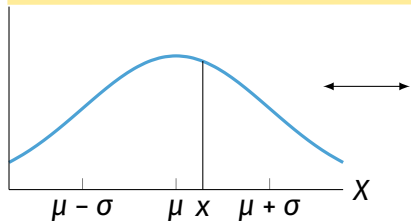
$$\sigma_X^2 = \int_{-\infty}^{+\infty} (x - \mu_X)^2 f_X(x) dx.$$

Note: these are essentially the same formulas as for discrete R.V.'s but with integrals instead of summations.

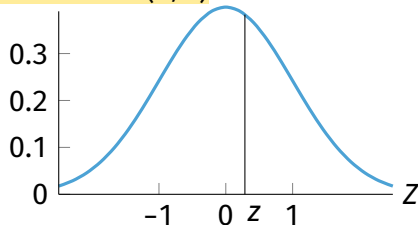
Note: you are **not** expected to know how to calculate integrals!

Standard Normal Distribution

normal distribution $x \in X \sim \text{Nor}(\mu, \sigma)$



standard normal distribution
 $z \in Z \sim \text{Nor}(0, 1)$

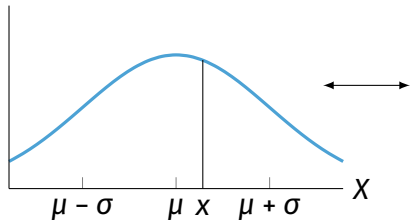


x and z have a similar position on the Gaussian bell curve.
What is the mathematical relationship between x and z ?

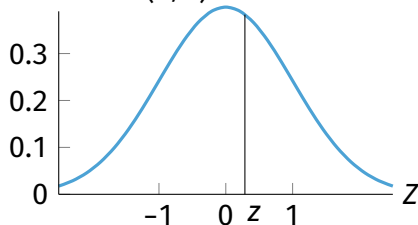
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Standard Normal Distribution

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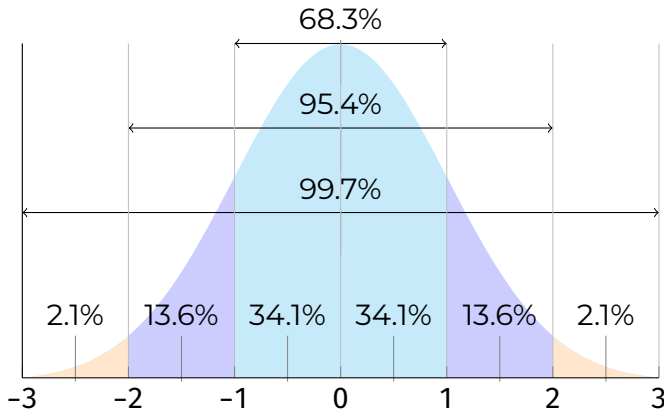


x and z have a similar position on the Gaussian bell curve.
What is the mathematical relationship between x and z ?

$$x = \mu + z \cdot \sigma \quad \text{and} \quad z = \frac{x - \mu}{\sigma}$$

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Standard Normal Distribution



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Python functions

Import `scipy.stats`

For a normal distribution with mean m and standard deviation s :

Function	Purpose
<code>norm.pdf(x, loc=m, scale=s)</code>	Probability density at x
<code>norm.cdf(x, loc=m, scale=s)</code>	Left-tail probability $P(X < x)$
<code>norm.sf(x, loc=m, scale=s)</code>	Right-tail probability $P(X > x)$
<code>norm.isf(1-p, loc=m, scale=s)</code>	$p\%$ of observations are expected to be lower than result

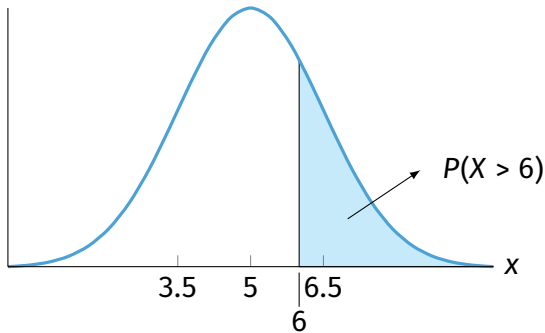
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Calculating Probabilities

What is the probability that Superman's observed reaction speed is over 6 ms?

Mathematical Notation:

$$P(X > 6) = ? \quad \text{with } X \sim \text{Nor}(\mu = 5, \sigma = 1,5)$$



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Calculating Probabilities: z-table

$P(X > 6) = ?$ with $X \sim \text{Nor}(\mu = 5, \sigma = 1, 5)$

(Old) calculation method using z-table, e.g.

<http://sixsigmastudyguide.com/wp-content/uploads/2014/04/z-table.jpg>

Calculating Probabilities: z-table

$P(X > 6) = ?$ with $X \sim \text{Nor}(\mu = 5, \sigma = 1, 5)$

(Old) calculation method using z-table, e.g.

<http://sixsigmastudyguide.com/wp-content/uploads/2014/04/z-table.jpg>

1. Calculate the z-score

$$z = \frac{6-5}{1,5} = 0,667 \text{ so } P(X > 6) = P(Z > 0,667)$$

2. Convert to a **left** tail probability

- o 100% probability rule: $P(Z > 0,667) = 1 - P(Z < 0,667)$
- o or symmetry rule: $P(Z > 0,667) = P(Z < -0,667)$

3. look for the corresponding value in the z-table

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Calculating Probabilities: using Python

$P(X > 6) = ?$ with $X \sim \text{Nor}(\mu = 5, \sigma = 1, 5)$

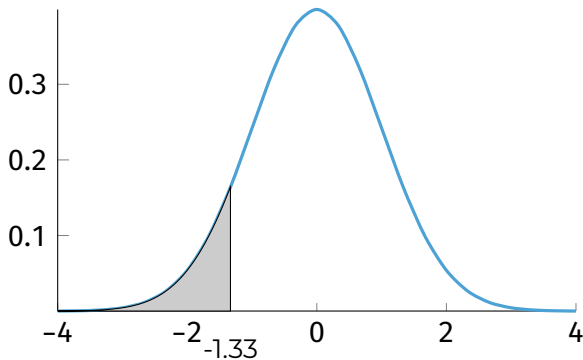
Calculated directly using Python:

```
import scipy.stats as stats
stats.norm.sf(6, loc=5, scale=1.5)
```

Examples

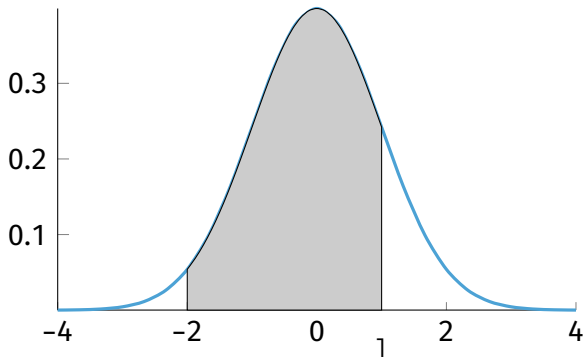
1. What is the probability that Superman reacts in less than 4 ms?
2. What is the probability that he reacts in less than 7 ms?
3. What is the probability that he reacts in less than 3 ms?
4. What is the probability that he reacts between 2 en 6.5 ms?
5. What interval contains 80% of his reaction speed?

Question 3: $P(X < 3)$



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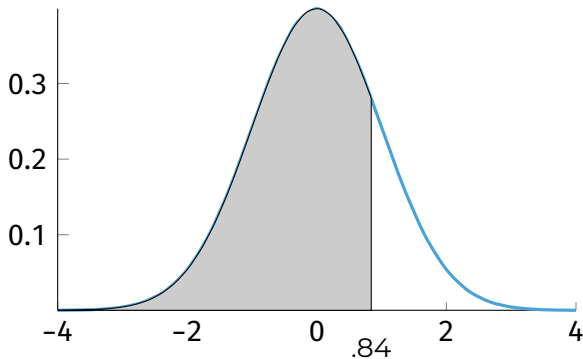
Question 4: $P(2 < X < 6, 5)$



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Question 5

For what value of x is $P(X < x) = 80\%$?



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Exponential Distribution

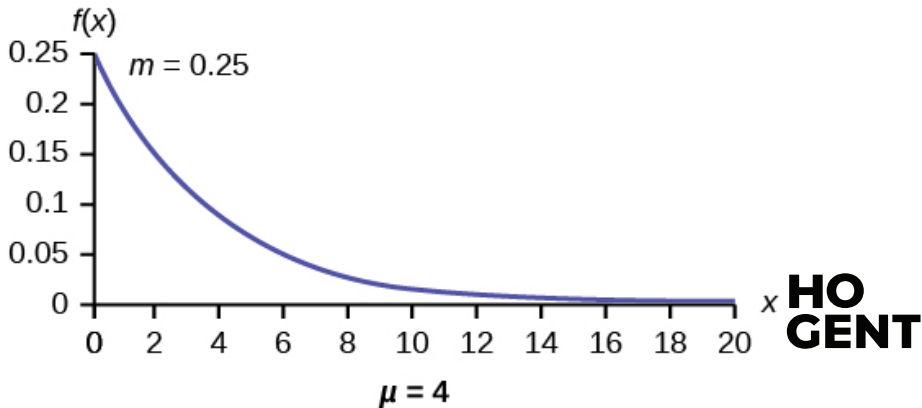
Besides the Normal Distribution, there are other often used continuous distributions, e.g. the **Exponential Distribution**.

Values for an exponential random variable occur when there are fewer large values and more small values. For example, the amount of money customers spend in a trip to the supermarket follows an exponential distribution. There are more people who spend small amounts of money and fewer people who spend large amounts of money.

Another examples is the length, in minutes, of long distance business telephone calls.

Exponential Distribution

Another example is the amount of time (in minutes) a postal clerk spends with his or her customers. In the image below, the time is known to have an exponential distribution with an average amount of time equal to four minutes.



Continuous Uniform Distribution

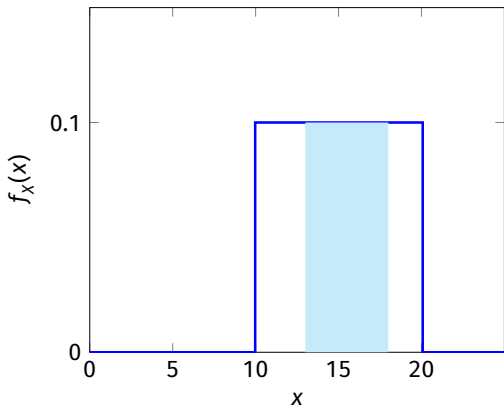
The continuous **Uniform Distribution** describes an experiment where there is an arbitrary outcome that lies between certain bounds.

It is the simplest of all continuous probability distributions. The density function is constant where every value has an equal chance of occurring.

Imagine you live in a building that has an elevator that will take you to your floor. From experience, once you push the button to call the elevator, it takes between ten and twenty seconds to arrive at your floor. This means the elevator arrival time is uniformly distributed between 10 and 20 seconds once you hit the button.

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Continuous Uniform Distribution



$$\begin{aligned}P(13 < X < 18) &= \text{area under } f_X(x) \text{ between 13 \& 18} \\&= 5 \times 0.1 \\&= 0.5\end{aligned}$$

Note:

$$\begin{aligned}P(13 < X < 18) &= P(13 < X \leq 18) = P(13 \leq X < 18) \\&= P(13 \leq X \leq 18)\end{aligned}$$

and it also holds that

$$P(10 \leq X \leq 20) = 1$$

so the total area under the probability density function is always equal to 1.

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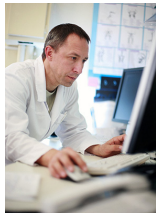
From Sample to Population

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The Central Limit Theorem

Central limit theorem

If the size of the sample is sufficiently large, the probability distribution of the sample mean will approximate a normal distribution, regardless of the probability distribution of the underlying population.



- 1 test
- 25 tests
- 100 tests



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Demo:

<https://www.zoology.ubc.ca/~whitlock/Kingfisher/CLT.htm>

The Central Limit Theorem

Consider a random sample of n observations drawn from a population with expected value μ and standard deviation σ . If n is sufficiently large, the probability distribution of the sample mean $\bar{x}$ will approximate a normal distribution with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

The larger the sample, the better the probability distribution of $\bar{x}$ will approximate the expected value of the population, μ .

Point Estimate

Point estimate

A **Point Estimate** for a population parameter is a formula or equation that allows us to calculate a value to estimate that parameter.

Example: sample variance/standard deviation, sample mean.

Confidence Interval

confidence interval

A **confidence interval** is an equation or formula that allows us to construct an interval that will contain the parameter to be estimated with a certain **level of confidence**.

Demo:

<https://www.zoology.ubc.ca/~whitlock/Kingfisher/CIMean.htm>

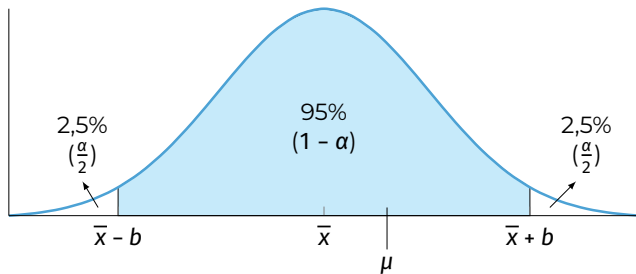
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Conf. Int. - Large Sample

Given a sample with mean $\bar{x}$.

We are looking for an interval $[\bar{x} - b, \bar{x} + b]$ for which we can say with a level of confidence $(1 - \alpha)$ of e.g. 95% that μ is inside this interval.

$$P(\bar{x} - b < \mu < \bar{x} + b) = 1 - \alpha = 0,95$$



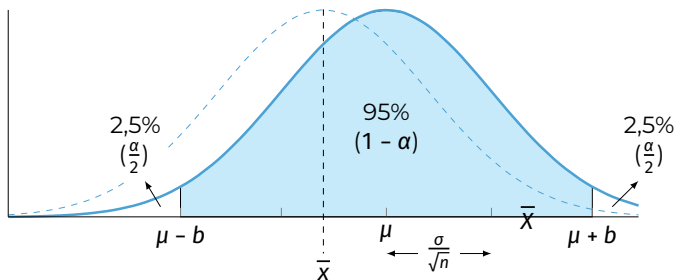
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Conf. Int. - Large Sample

Because of the central limit theorem we know that: $\bar{X} \in \bar{X} \sim \text{Nor}(\mu, \frac{\sigma}{\sqrt{n}})$

And because of the symmetry we can say:

$$P(\bar{X} - b < \mu < \bar{X} + b) = P(\mu - b < \bar{X} < \mu + b)$$



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Conf. Int. - Large Sample

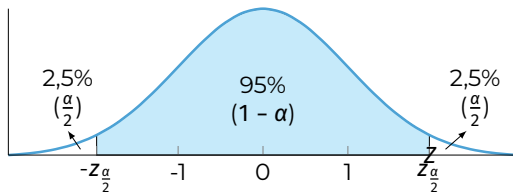
We now calculate the z-score for $\bar{x}$: $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

We lookup (or calculate) the value $z_{\frac{\alpha}{2}}$ for which:

$$P\left(-z_{\frac{\alpha}{2}} < z < z_{\frac{\alpha}{2}}\right) = 1 - \alpha = 0.95$$

$$P\left(z < z_{\frac{\alpha}{2}}\right) = 1 - \frac{\alpha}{2} = 0.975$$

$$z_{\frac{\alpha}{2}} = \text{stats.norm.isf}(1 - 0.975) \approx 1.96$$



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Conf. Int. - Large Sample

The result is:

$$P\left(-1,96 < \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} < 1,96\right) = 0,95$$

$$P\left(\mu - 1,96 \frac{\sigma}{\sqrt{n}} < \bar{x} < \mu + 1,96 \frac{\sigma}{\sqrt{n}}\right) = 0,95$$

Because of symmetry we can swap μ and $\bar{x}$:

$$P\left(\bar{x} - 1,96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1,96 \frac{\sigma}{\sqrt{n}}\right) = 0,95$$

Now we can say with 95% confidence that:

$$\mu \in \left[\bar{x} - 1,96 \cdot \frac{\sigma}{\sqrt{n}} , \bar{x} + 1,96 \cdot \frac{\sigma}{\sqrt{n}} \right]$$

59/63 (in practice we will use s_{sample} as a point estimate for $\sigma_{population}$)

Conf. Int. - Small Sample

For a ***small*** sample the central limit theorem is **no longer** valid.

Instead we can say:

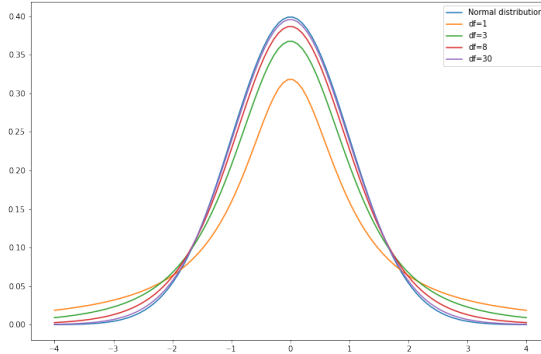
If a population X has a normal distribution ($X \sim \text{Nor}(\mu, \sigma)$) and you take a *small* sample with mean $\bar{x}$ and standard deviation s , then

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

will behave as a t-distribution with $n - 1$ degrees of freedom

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Student t-distribution



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Student t -distribution in Python

```
Import scipy.stats
```

For a t -distribution with df degrees of freedom:

(df = degrees of freedom)

Funcție	Betekenis
<code>t.pdf(x, df=d)</code>	Probability density for x
<code>t.cdf(x, df=d)</code>	Left-tail probability $P(X < x)$
<code>t.sf(x, df=d)</code>	Right-tail probability $P(X > x)$
<code>t.isf(1-p, df=d)</code>	$p\%$ of observations is expected to be lower than this value

Conf. Int.: Small Sample

To determine the confidence interval for the mean μ of a population, based on a *small* sample, we first calculate $t_{\frac{\alpha}{2}, df}$.

With a confidence level of 95% we have $\frac{\alpha}{2} = 0,025$
Assume for example $n = 5$ (so $df=4$), then we have

$$t_{\frac{\alpha}{2}, df} = \text{stats.t.isf}(1-0.975, df=4) = 2.776$$

We can say with a confidence of 95% that:

$$\mu \in \left[\bar{x} - t_{\frac{\alpha}{2}, df} \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{\frac{\alpha}{2}, df} \cdot \frac{s}{\sqrt{n}} \right]$$

Module 3b. Hypothesis testing

Data Science & AI

Sabine De Vreese Lieven Smits Bert Van Vreckem
2023–2024

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Contents

Testing procedure

Probability Value

Critical Region

Examples

Student's t-test

Errors in Hypothesis Tests

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Learning Goals

- Statistical hypothesis testing concepts
- Hypothesis testing procedure
- Apply the z - and t -test

Testing procedure

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Statistical Hypothesis Testing

Hypothesis Idea that has yet to be proven: statement regarding the numeric value of a population parameter

Hypothesis Test verification of a statement about the values of one or multiple population parameters

Null Hypothesis (H_0) Base hypothesis, we start with assuming it is true

Alternative Hypothesis (H_1 , H_a) Conclusion if the null hypothesis is unlikely to be true

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Elements of a testing procedure

Test Statistic The value that is calculated from the sample

Region of Acceptance The region of values **supporting** the null hypothesis

Critical Region / Region of Rejection The region of values **rejecting** the null hypothesis

Significance Level The probability of rejecting a true null hypothesis H_0

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Testing procedure

1. Formulate both hypotheses (H_0 and H_1)
2. Determine the significance level (α)
3. Calculate the test statistic
4. Determine the critical region or the probability value
5. Draw conclusions

Hypotheses about superheroes



A superhero rescues 3.3 persons a day



Source: <http://www.cracked.com/quick-fixes/4-people-who-saved-day-while-dressed-as-superheroes/>

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Assume that, over a period of 30 days, on average 3.483 people were saved each day ($\bar{x} = 3.483$, $n = 30$)

1. Hypothesis: $H_0 : \mu = 3.3$; $H_1 : \mu > 3.3$
2. Significance level: $\alpha = 0.05$
3. Test statistic: $\bar{x} = 3.483$

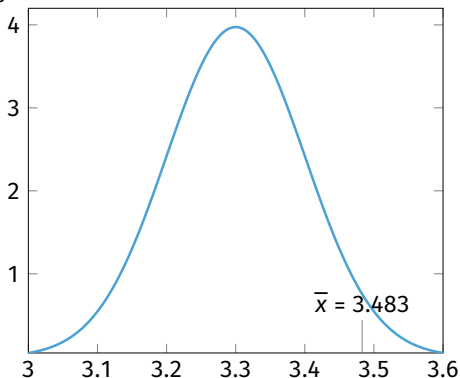
Standard deviation of the population (assumed to be known): $\sigma = 0.55$

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Calculate test statistic

Based on the central limit theorem, we know that:

$$M \sim \text{Nor}(\mu = 3.3; \sigma = \frac{0.55}{\sqrt{30}} = 0.1)$$



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Probability Value

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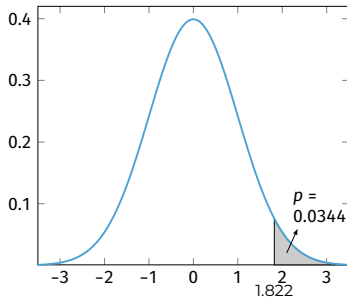
Probability Value

p -value

The p -value is the probability, if the null hypothesis is true, to obtain a value for the test statistic that is at least as extreme as the observed value.

- $p\text{-value} < \alpha \Rightarrow$ reject H_0 : the discovered value of $\bar{x}$ is too extreme;
- $p\text{-value} \geq \alpha \Rightarrow$ do not reject H_0 : the discovered value of $\bar{x}$ can still be explained by coincidence.

Probability Value



$$P(M > 3.483) = P\left(Z > \frac{3.483 - 3.3}{\frac{\sigma}{\sqrt{n}}}\right) = P(Z > 1.822) = 0.0344$$

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Critical Region

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Critical Region

Critical region

The **critical region** is the collection of all values of the test statistic for which we can reject the null hypothesis.

We look for a critical value g for which:

$$P(M > g) = \alpha$$

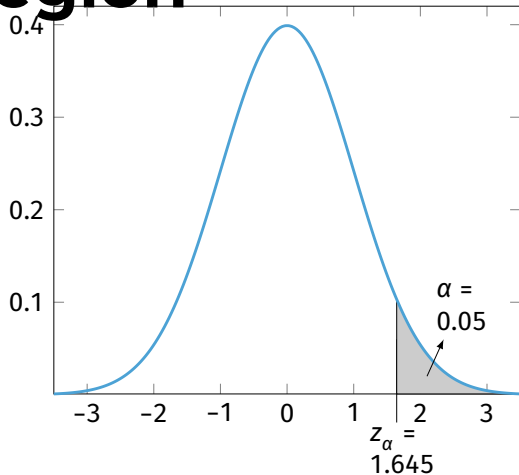
Determine z_α for which:

$$P(Z > z_\alpha) = \alpha \Rightarrow g = \mu + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$$

- Left of g : region of acceptance (do not reject H_0)
- Right of g : critical region (reject H_0)

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Critical Region



significance level $\alpha = 0.05 \Rightarrow$ critical value $z_\alpha = 1.645$
($z_\alpha = \text{stats.norm.isf}(1 - 0.95)$)

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Left-tailed testing

What would you have to change in the equation in order to calculate the correct critical value?

Left-tailed testing

What would you have to change in the equation in order to calculate the correct critical value? Answer:

$$g = \mu - z \times \frac{\sigma}{\sqrt{n}}$$

because

$$P(M < g) = P\left(Z < \frac{g - \mu}{\frac{\sigma}{\sqrt{n}}}\right) = 0.05$$

Left-tailed testing

Because of symmetry:

$$P\left(Z > -\left(\frac{g - \mu}{\frac{\sigma}{\sqrt{n}}}\right)\right) = 0.05$$

The corresponding z-value is 1.645, and therefore:

$$\begin{aligned} z = \frac{-g + \mu}{\frac{\sigma}{\sqrt{n}}} &\Leftrightarrow -g = \frac{\sigma}{\sqrt{n}} z_{\alpha} - \mu \\ &\Leftrightarrow g = \mu - z_{\alpha} \frac{\sigma}{\sqrt{n}} \end{aligned}$$

Two-tailed testing

Sometimes it can be necessary to perform a two-tailed test. In this case, two critical values need to be calculated, namely the left and right critical value.

$$g = \mu \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \quad (1)$$

Summary

Goal	Test regarding the value of the population mean μ using a sample of n independent values		
Prerequisite	De population has a random distribution, n is sufficiently large		
Test Type	Two-tailed	Left-tailed	Right-tailed
H_0	$\mu = \mu_0$	$\mu = \mu_0$	$\mu = \mu_0$
H_1	$\mu \neq \mu_0$	$\mu < \mu_0$	$\mu > \mu_0$
Critical Region	$ \bar{x} > g$	$\bar{x} < -g$	$\bar{x} > g$
Test statistic	$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$		

Table: Summary of Testing Procedures

Requirements for z-test

- The sample needs to be random
- The sample size needs to be sufficiently large ($n \geq 30$)
- The test statistic needs to have a normal distribution
- The standard deviation of the population, σ , is known

Sometimes these assumptions will not hold and in this case we can *not* use the Z-test!

Examples

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Example 1

When drawing a random sample consisting of 50 observations from a population with variance $\sigma^2 = 55$ we obtain as sample mean $\bar{x} = 25$. We now want to find out if there is a reason to assume that the population mean is smaller than 27.

Example 1

- 1 Formulate both hypotheses
 $H_0 : \mu = 27$ en $H_1 : \mu < 27$.
- 2 Determine significance level α and sample size n
 $\alpha = 0.05$ en $n = 50$
- 3 Test statistic & value: sample mean $\bar{x} = 25$

Example 1 (cont.)

4a Probability Value

According to the central limit theorem:

$$M \sim \text{Nor}(\mu = 27, \frac{\sigma^2}{\sqrt{n}})$$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{25 - 27}{\sqrt{\frac{55}{50}}} \approx -1.91$$

We therefore have a probability value of 0.0281.

Using a significance level of 0.05, we can reject H_0 .

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Example 1 (cont.)

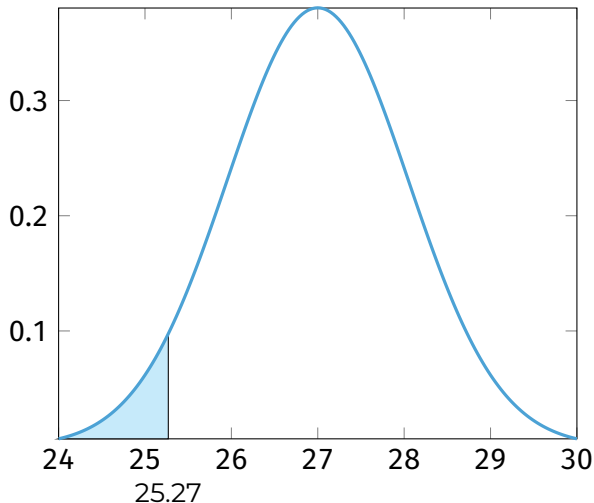
4b Calculate and plot the critical region

$$\begin{aligned}g &= \mu - z \times \frac{\sigma}{\sqrt{n}} \\&= 27 - 1.645 \times \sqrt{\frac{\sigma}{n}} \\&= 25.27470944\end{aligned}$$

We have that $\bar{x} < g$ and therefore we can reject H_0 .

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Example 1 (cont.)



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Example 1 (cont.)

5 Conclusion

We can conclude, based on the sample, that $\mu < 27$ for a significance level of 0.05

Example 2

In a research about the amount of change in the pockets of our superheroes, researchers state that on average a superhero carries €25 of cash. We assume that the standard deviation of the population $\sigma = 7$. For a random sample of size $n = 64$, the average amount of money a superhero carries is $\bar{x} = €23$. For the significance level, $\alpha = 0.05$ is selected.

Example 2

- 1 Formulate both hypotheses
 $H_0 : \mu = 25$ en $H_1 : \mu \neq 25$
- 2 Determine significance level α and sample size n
 $\alpha = 0.05$ en $n = 64$.
- 3 Test statistic & value: $\bar{x} = 23$

Example 2 (cont.)

4b Calculate the critical region

$$g_1 = \mu - z \times \frac{\sigma}{\sqrt{n}} = 23.28$$

$$g_2 = \mu + z \times \frac{\sigma}{\sqrt{n}} = 26.72$$

We have that $\bar{x}$ is inside the critical region (because $23 < 23.28$) so we can reject H_0 .

5 Based on this sample we can conclude that superheroes carry on average *less* than 25 €, using a significance level of 5%

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Student's t-test

Student's t-test

What if the requirements for a z-test are not met? E.g.

- Sample size too small
- Population stdev (σ) unknown

If the variable is normally distributed, you can use the *t*-test

The *t*-test

Determine critical value:

$$g = \mu \pm t \times \frac{s}{\sqrt{n}}$$

- *t*-value is derived from the Student's *t*-distribution, based on the number of *degrees of freedom*, $n - 1$
- Look for value using the function `t.isf()` in Python
- Apart from this, the procedure is identical to the procedure of the *z*-test

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Errors in Hypothesis Tests

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Errors in Hypothesis Tests

Conclusion	Reality	
	H_0 True	H_1 True
H_0 not rejected	Correct inference	Type II error (false negative)
H_0 rejected	Type I error (false positive)	Correct inference

$P(\text{type I error}) = \alpha$ (= significance level)

$P(\text{type II error}) = \beta$

Calculating β is **not** trivial, but if $\alpha \searrow$ then $\beta \nearrow$

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Module 4. Bivariate analysis: qualitative variables

Data Science & AI

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2023–2024

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Contents

Bivariate analysis

Contingency tables

- The chi-squared statistic

- Cramér's V

Chi-squared test for independence

Goodness-of-fit test

- Testing procedure for goodness of fit test

Standardized residuals

Cochran's rules

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Learning goals

- Dependent/independent variable
- Apply suitable analysis techniques for each combination of measurement levels
- Contingency tables and Cramér's V
- Visualization

Bivariate analysis: overview

Independent	Dependent	Test/Metric
Qualitative	Qualitative	χ^2 -test Cramér's V
Qualitative	Quantitative	two-sample t -test Cohen's d
Quantitative	Quantitative	— Regression, correlation

Bivariate analysis

Bivariate analysis

- ...is determining whether there is an association between two stochastic variables (X and Y).
- **Association** = you can predict (to some extent) the value of Y from the value of X
 - X — Independent variable
 - Y — Dependent variable
- **Important!** Finding an association does **NOT** imply a causal relation!

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Example research questions

- Is there a difference in taste preference between two beverage brands?
- Is there a difference in spending at the campus restaurant between students and staff?
- Do smokers die more often of lung cancer than non-smokers?
- Do men and women have a different opinion on a survey question?
- ...

We will use `data/rlanders.csv` from the Github repo for lab assignments to explore the last question.

Contingency tables

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Contingency tables

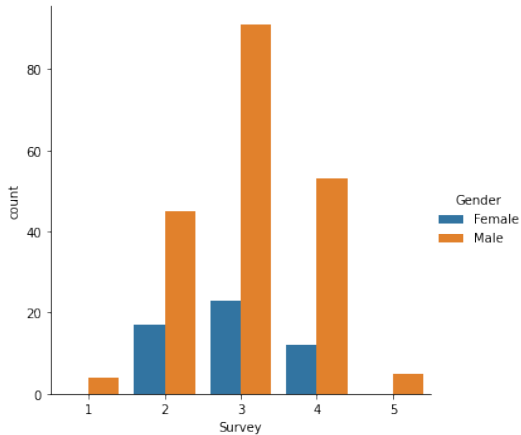
(also: crosstab)

See Python example code in `demo-chi-squared`

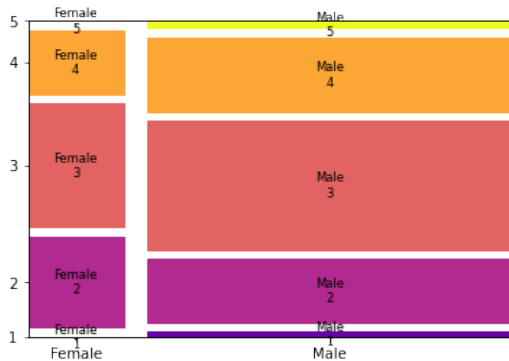
Gender Survey	Female	Male
Strongly disagree	0	4
Disagree	17	45
Neutral	23	91
Agree	12	53
Strongly agree	0	5

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Visualization



A clustered bar chart



A mosaic plot

Marginal totals

Gender Survey	Female	Male	Total
Strongly disagree	0	4	4
Disagree	17	45	62
Neutral	23	91	114
Agree	12	53	65
Strongly agree	0	5	5
Total	52	198	250

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Expected values

If there is no difference (association), we expect the same ratios in each column of the table!

Gender Survey	Female	Male	Total
Strongly disagree	0.832	3.168	4
Disagree	12.896	49.104	62
Neutral	23.712	90.288	114
Agree	13.520	51.480	65
Strongly agree	1.040	3.960	5
Total	52	198	250

In each cell: (row total $\times$ column total) / n

Measuring dispersion

How far is the observed value o from the expected e ?

$$\frac{(o - e)^2}{e}$$

Gender Survey	Female	Male
Strongly disagree	0.832	0.219
Disagree	1.306	0.343
Neutral	0.021	0.006
Agree	0.171	0.045
Strongly agree	1.040	0.273

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The chi-squared statistic

The sum of all these values is notated:

$$\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i} \approx 4.255$$

- χ is the Greek letter *chi*
- o_i = number of observations in the i 'th cell of the contingency table
- e_i = expected frequency
- Small value $\Rightarrow$ no association
- Large value $\Rightarrow$ association

When is χ^2 large enough?

- 2×2 -table with $\chi^2 = 10$
 - Relatively large difference
 - Indicates association
- 5×5 -table with $\chi^2 = 10$
 - Relatively small difference
 - Does NOT indicate association

We need a metric independent of table size!

Cramér's V

$$V = \sqrt{\frac{\chi^2}{n(k-1)}} = \sqrt{\frac{4.255}{250(2-1)}} \approx 0.130$$

with n the number of observations, $k = \min(\text{numRows}, \text{numCols})$

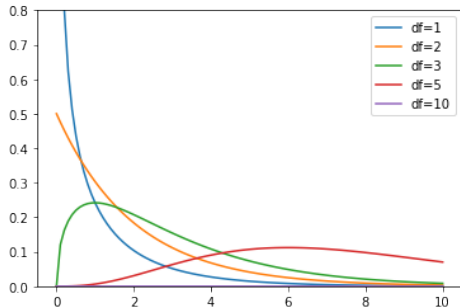
Cramér's V	Interpretation
≈ 0	no association
≈ 0.1	weak association
≈ 0.25	moderate association
≈ 0.5	strong association
≈ 0.75	very strong association
≈ 1	complete association

Chi-squared test for independence

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χ^2 test for independence

- = Alternative to Cramér's V to investigate association between qualitative variables.
- Value of χ^2 distributed according to the χ^2 distribution



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χ^2 -distribution in Python

Import `scipy.stats`

For a χ^2 -distribution with `df` degrees of freedom:

Function	Purpose
<code>chi2.pdf(x, df=d)</code>	Probability density for <code>x</code>
<code>chi2.cdf(x, df=d)</code>	Left-tail probability $P(X < x)$
<code>chi2.sf(x, df=d)</code>	Right-tail probability $P(X > x)$
<code>chi2.isf(1-p, df=d)</code>	<code>p%</code> of observations is expected to be lower than this value

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Test procedure

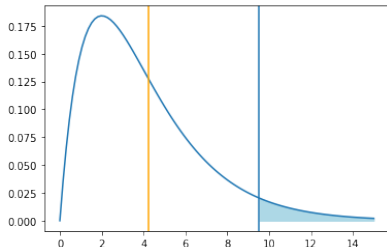
- **Step 1.** Formulate hypotheses:
 - H_0 : there is no association (χ^2 is “small”)
 - H_1 : there is an association (χ^2 is “large”)
- **Step 2.** Choose significance level, e.g. $\alpha = 0.05$
- **Step 3.** Calculate the test statistic, $\chi^2 = 4.255$

Test procedure (cont.)

- **Step 4.** Use $df = (numRow - 1) \times (numCol - 1)$ and either:
 - Determine critical value g so $P(\chi^2 > g) = \alpha$
 - Calculate the p -value
- **Step 5.** Draw conclusion:
 - $\chi^2 < g$: do not reject H_0 ; $\chi^2 > g$: reject H_0
 - $p > \alpha$: do not reject H_0 ; $p < \alpha$: reject H_0

Example (Gender vs Survey)

- `g = stats.chi2.isf(0.05, df=4)` (result: 9.4877)
- `p = stats.chi2.sf(4.2555, df=4)` (result: 0.3725)



Conclusion: we do not reject the null hypothesis. Differences between expected and observed values are not significantly large.
We found no association between variables *Gender* and *Survey*

Test for independence in Python

SciPy has a function to calculate χ^2 and p -value from a contingency table:

```
1 observed = pd.crosstab(rlanders.Survey, rlanders.Gender)
2 chi2, p, df, expected = stats.chi2_contingency(observed)
3
4 print("Chi-squared           : %.4f" % chi2)
5 print("Degrees of freedom: %d" % df)
6 print("P-value             : %.4f" % p)
```

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Goodness-of-fit test

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Goodness-of-fit test

The χ^2 test can also be used to determine whether a sample is **representative** for the population.

Goodness-of-fit test

This test indicates to what degree a sample corresponds to a null hypothesis regarding the distribution of a qualitative variable over mutually exclusive classes.

Example



Type	# sample	# population
Mutant	127	35%
Human	75	17%
Alien	98	23%
God	27	8%
Demon	73	17%
Total	400	100%

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Example

Is the distribution of the sample ($n = 400$) representative for the full population (all superheroes)?

- What numbers would you *expect* if the sample is representative?
- How large are the differences from the *observed* numbers?
 - small $\Rightarrow$ distribution is representative
 - large $\Rightarrow$ distribution is **not** representative

Example

Is the distribution of the sample ($n = 400$) representative for the full population (all superheroes)?

- What numbers would you *expect* if the sample is representative?
- How large are the differences from the *observed* numbers?
 - small $\Rightarrow$ distribution is representative
 - large $\Rightarrow$ distribution is **not** representative

Can you see the link with contingency tables and Cramer's V?

Goodness of fit test

- Exactly representative $\Rightarrow$ 35% of superheroes in the sample is a mutant
- The expected number therefore is $e = 0.35 \times 400 = 140$.

Therefore:

$$e = n \times \pi$$

If the differences $o - e$ are relatively small they can be attributed to random sampling errors.

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Goodness of fit test

Consider χ^2 :

$$\chi^2 = \sum_{i=1}^n \frac{(o_i - e_i)^2}{e_i}$$

Draw a conclusion based on the value of χ^2 :

- small $\Rightarrow$ distribution is representative
- large $\Rightarrow$ distribution is **not** representative

χ^2 measures the degree of conflict with the null hypothesis

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Goodness of fit test

Superhero type	o	π	e	$\frac{(o-e)^2}{e}$
Mutant	127	35%	140	1.21
Human	75	17%	68	0.72
Alien	98	23%	92	0.39
God	27	8%	32	0.78
Demon	73	17%	68	0.37
			$\chi^2 =$	3.47

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Goodness of fit test

- The test statistic χ^2 follows the χ^2 distribution.
- Critical value g from the χ^2 distribution: this is dependent on the number of degrees of freedom (df). In general:

$$df = k - 1$$

with k the number of categories.

- The critical value g for a given significance level α and number of degrees of freedom df can be calculated in Python using the function `isf()`.

$$P(\chi^2 < g) = 1 - \alpha$$

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Goodness of fit test

Testing Procedure

1. Formulate hypotheses

- o H_0 : sample is representative for the population
- o H_1 : sample is not representative for the population

2. Choose significance level: $\alpha = 0.05$

Goodness of fit test

Testing Procedure

1. Calculate test statistic:

$$\chi^2 = \sum_{i=1}^n \frac{(o_i - e_i)^2}{e_i}$$

1.1 **Critical area:** Calculate g so that $P(\chi^2 < g) = 1 - \alpha$

1.2 **Probability value:** Calculate $p = 1 - P(X < \chi^2)$

2. Conclusion (the test is always right-tailed):

2.1 $\chi^2 < g \Rightarrow$ do not reject H_0 , $\chi^2 > g \Rightarrow$ reject H_0

2.2 $p > \alpha \Rightarrow$ do not reject H_0 , $p < \alpha \Rightarrow$ reject H_0

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Example

- `g = stats.chi2.isf(0.05, df=4)` (result: 9.4877)
- `p = stats.chi2.sf(3.4679, df=4)` (result: 0.4828)

Conclusion. $\chi^2 \approx 3.47 < g \approx 9.4877$, so we don't reject the null hypothesis.
This sample is representative for the population.

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Goodness-of-fit test in Python

```
1 observed = np.array([127, 75, 98, 27, 73])
2 expected_p = np.array([.35, .17, .23, .08, .17])
3 expected = expected_p * sum(observed)
4 chi2, p = stats.chisquare(f_obs=observed, f_exp=expected)
5
6 print("χ² = %.4f" % chi2)
7 print("p = %.4f" % p)
```

Standardized residuals

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Example: families

Consider all families with exactly 5 children in a given community.

Example: families

Consider all families with exactly 5 children in a given community. When we look at the number of boys/girls, there are 6 possible combinations:

1. 5 boys
2. 4 boys, 1 girl
3. 3 boys, 2 girls
4. 2 boys, 3 girls
5. 1 boy, 4 girls
6. 5 girls

A survey was conducted regarding 1022 families with exactly 5 children

Are the observed numbers in the 6 classes representative for a population in which the probability of having a boy is equal to the probability of having a girl, or more concrete 0.5?

Example

i	0	1	2	3	4	5
o_i	58	149	305	303	162	45

Example

i	0	1	2	3	4	5
o_i	58	149	305	303	162	45

If the assumption is true, the probability π_i to have i boys is determined by a binomial distribution with parameters $n = 5$ and $p = 0.5$. For example, the probability to have 2 boys out of 5 children is equal to:

$$(0.5)^2 \times (1 - 0.5)^{5-2} \times \binom{5}{2}$$

In general (you don't have to know why):

$$\pi_i = \binom{5}{i} \times 0.5^i \times 0.5^{5-i} = \frac{5!}{i!(5-i)!} \times 0.5^i$$

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Example

i	0	1	2	3	4	5	Total
o_i	58	149	305	303	162	45	1022
π_i	0,031	0,156	0,313	0,313	0,156	0,031	1
e_i	31,9	159,7	319,4	319,4	159,7	31,9	1022
$\frac{(o-e)^2}{e}$	21,268	0,715	0,647	0,840	0,033	5,343	28,846
r_i	4,686	-0,921	-0,970	-1,105	0,199	2,348	

Example

1. **Formulate both hypotheses**

- o H_0 : the sample is representative for the population
- o H_1 : the sample is not representative for the population

2. **Determine α and n : $\alpha = 0.01$ and $n = 1022$.**

3. **Value of the test statistic in the sample:**

$$\chi^2 = \sum_{i=1}^n \frac{(o_i - e_i)^2}{e_i} \approx 28.846$$

4. **Calculate and plot critical area:** The critical value is 15.0863. Our test statistic is inside the critical area, so we can reject H_0 .

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Standardized Residuals

Standardized Residuals

The standardized residuals indicate which classes make the greatest contribution to the value of χ^2 .

$$r_i = \frac{o_i - n\pi_i}{\sqrt{n\pi_i(1 - \pi_i)}}$$

- $r_i \in [-2, 2] \Rightarrow$ “acceptable” values
- $r_i < -2 \Rightarrow$ underrepresented
- $r_i > 2 \Rightarrow$ overrepresented

Conclusion: families in which all children are of the same gender are overrepresented.

Cochran's rules

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Conditions

In order to apply the χ^2 -test, the following conditions must be met (Rule of Cochran)

1. For all categories, the expected frequency e must be greater than 1.
2. In a maximum of 20 % of the categories, the expected frequency e may be less than 5.

Module 5. Bivariate analysis: qualitative - quantitative

Data Science & AI

Sabine De Vreese Lieven Smits Bert Van Vreckem
2023–2024

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Contents

Data visualization

Two-sample t-test

Paired samples

Effect size

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Learning goals

- Apply the t -test for two samples
- Calculate effect size using Cohen's d
- Visualization

Bivariate analysis: overview

Independent	Dependent	Test/Metric
Qualitative	Qualitative	χ^2 -test Cramér's V
Qualitative	Quantitative	two-sample t -test Cohen's d
Quantitative	Quantitative	— Regression, correlation

Example research questions

- Are male penguins larger than females?
- Do men get a higher salary than women?
- Does a new vaccine protect against a disease?
- Does “retrieval practice” improve learning outcomes (i.e. student grades)?
- ...

In these examples, what is the independent/dependent variable?

Data visualization

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Data visualization

- Chart types for quantitative data
- Grouped by qualitative variable

Data visualization

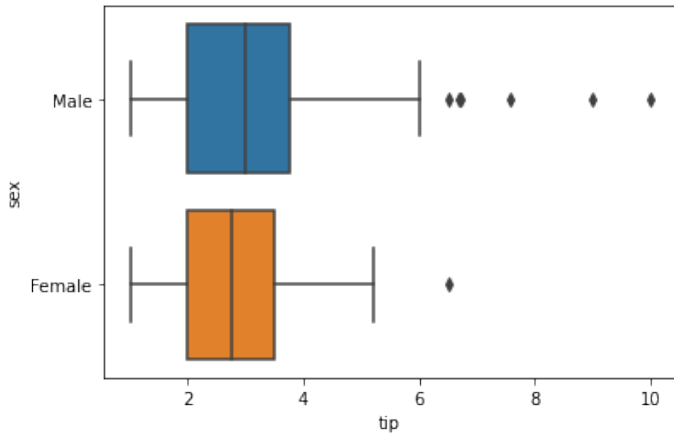
- Chart types for quantitative data
- Grouped by qualitative variable

Suitable chart types:

- Grouped boxplot
- Grouped density plot
- Bar chart with error bars
- ...

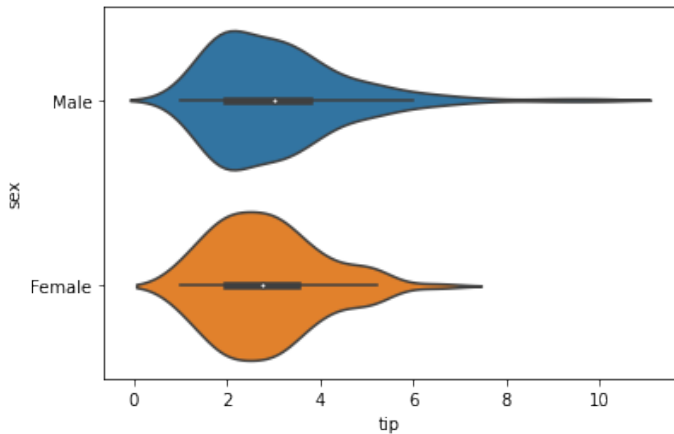
Grouped boxplot

(source code: see `demo-2sample-t.ipynb`)



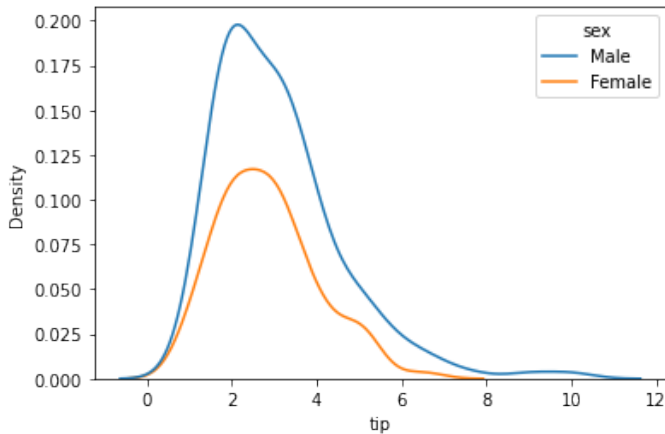
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Grouped violin plot



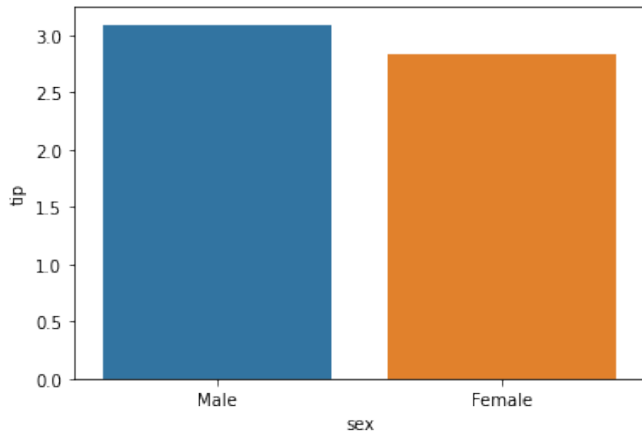
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Grouped density plot



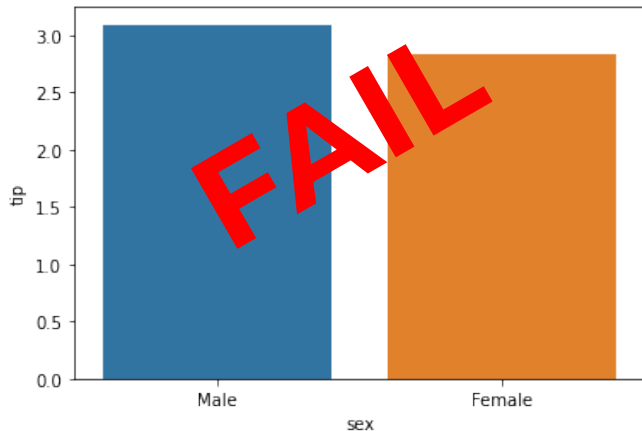
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Beware! Bar chart of group means



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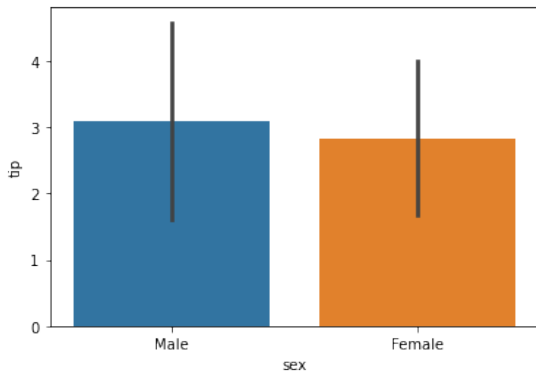
Beware! Bar chart of group means



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Bar chart with error bars

- Always add error bars!
- Only makes sense for normally distributed data
- Example: 1 standard deviation:



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Two-sample t-test

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Comparing two population means

Are the population means of two populations different?

We use two *samples* to perform an appropriate statistical test.

Correct test depends on:

- Independent samples
- Paired samples

Independent samples

In a clinical study, the aim is to determine whether a new drug has a delayed (i.e. higher) reaction time as a side effect.

- Control group: 6 participants receive placebo
- Intervention group: 6 participants receive medicine

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Independent samples

In a clinical study, the aim is to determine whether a new drug has a delayed (i.e. higher) reaction time as a side effect.

- Control group: 6 participants receive placebo
- Intervention group: 6 participants receive medicine

Next, the reaction time (in ms) is measured:

- Control group: 91, 87, 99, 77, 88, 91 ($\bar{x} = 88.83$)
- Intervention group: 101, 110, 103, 93, 99, 104 ($\bar{y} = 101.67$)

Are there significant differences between the intervention and control group?

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Independent samples

Testing procedure

1. Hypotheses:

- $H_0: \mu_1 - \mu_2 = 0$
- $H_1: \mu_1 - \mu_2 < 0$

2. Significance level: $\alpha = 0.05$

3. Test statistic is based on:

- $\bar{x} - \bar{y} = -12.833$
- $\bar{x}$ = estimation for μ_1 (control group)
- $\bar{y}$ = estimation for μ_2 (intervention group)
- Takes the variances of the samples into account. For completeness the test statistic is

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{s_x^2/n_x + s_y^2/n_y}}.$$

4. Calculate p

Independent samples

Calculation in Python

The calculation of the test statistic and the associated p -value is done by `stats.ttest_ind()`

```
control = np.array([91, 87, 99, 77, 88, 91])
treatment = np.array([101, 110, 103, 93, 99, 104])
stats.ttest_ind(a=control, b=treatment,
                alternative='less', equal_var=False)
```

Result:

```
Ttest_indResult(statistic=-3.445612673536487,
                pvalue=0.003391230079206901)
```

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Independent samples

Testing procedure (continued)

5. Draw conclusion based on p -value.

$p \approx 0.00339 < \alpha = 0.05$. We reject the null hypothesis. In this sample, there is reason to assume that the drug does indeed increase reaction time.

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Paired samples

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Paired samples

A study examined whether cars that run on petrol with additives have a lower consumption, i.e. a higher miles per gallon.

For 10 cars, the consumption was measured (expressed in miles per gallon) for both fuel types:

Car	1	2	3	4	5	6	7	8	9	10
Regular petrol	16	20	21	22	23	22	27	25	27	28
With additives	19	22	24	24	25	25	25	26	28	32

Paired samples

Testing procedure

1. Hypotheses:

o $H_0: \mu_{X-Y} = 0$

o $H_1: \mu_{X-Y} < 0$

2. Significance level: $\alpha = 0.05$

3. Test statistic:

o Based on the mean of the difference $\bar{d} = \overline{x - y}$

Regular petrol	16	20	21	22	23	22	27	25	27	28
With additives	19	22	24	24	25	25	25	26	28	32
Difference	-3	-2	-3	-2	-2	-3	2	-1	-1	-4

o For completeness, the test statistic is

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

4. Calculate p

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Paired samples

Calculation in Python

```
regular = np.array([16, 20, 21, 22, 23, 22, 27, 25, 27, 28])  
additives = np.array([19, 22, 24, 24, 25, 25, 26, 26, 28, 32])  
stats.ttest_rel(regular, additives, alternative='less')
```

Result:

```
Ttest_relResult(statistic=-4.47213595499958,  
                 pvalue=0.00077494295585091)
```

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Paired samples

5. Draw conclusion based on p -value.
 $p \approx 0.0007749 < \alpha = 0.05$. We reject the null hypothesis. In this sample, there is reason to assume that the fuel with additives leads to lower fuel consumption, i.e. higher miles per gallon.

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Effect size

Effect size

Effect size

The **effect size** is a metric which expresses how great the difference between two groups is

- Control group vs. intervention group
- Can be used in addition to hypothesis test
- Often used in educational sciences
- There are several definitions, here: *Cohen's d*

Cohen's d

$$d = \frac{\overline{x}_1 - \overline{x}_2}{s}$$

where $\overline{x}_1$ and $\overline{x}_2$ represent the sample means and s the pooled standard deviation, i.e. standard deviation of both groups combined:

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

with n_1, n_2 the sample sizes, and s_1, s_2 the standard deviations of both groups

Interpretation Cohen's d

$ d $	Effect Size
0.01	Very small
0.2	Small
0.5	Average
0.8	Large
1.2	Very large
2.0	Huge

In educational sciences (John Hattie):

- 0,4 = tipping point for desired effects
- effect size $d = 1$: process material that would normally take 1 year in 6 months!

E.g. <https://visible-learning.org/backup-hattie-ranking-256-effects-2017/>

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Typical approach research in education

- Research question: Is X a good learning strategy, in other words, does this have a positive effect on final results?
- Control group uses “traditional” approach
- Intervention group uses X
- Followed by an evaluation moment
- Determine scores, calculate d

Module 6. Bivariate analysis: quantitative—quantitative

Data Science & AI

Sabine De Vreese Lieven Smits Bert Van Vreckem
2023–2024

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Contents

Data visualization

Linear regression

Covariance

Pearson's correlation coefficient

Coefficient of determination

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Learning goals

- Determine the equation of the regression line and plot it;
- Calculate the covariance Cov , the correlation coefficient R and the coefficient of determination R^2
- Interpret these values using the correct terms;
- Visualization

Bivariate analysis: overview

Independent	Dependent	Test/Metric
Qualitative	Qualitative	χ^2 -test Cramér's V
Qualitative	Quantitative	two-sample t -test Cohen's d
Quantitative	Quantitative	— Regression, correlation

Data visualization

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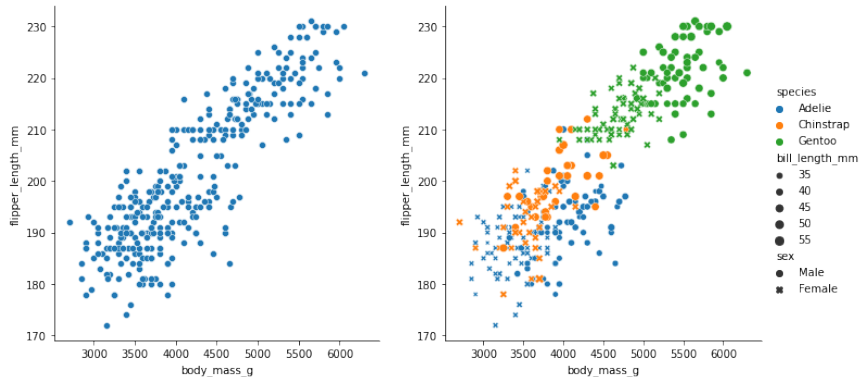
Data visualization

To visualize quantitative data, we use a *scatter plot*

- X-axis: independent variable
- Y-axis: dependent variable
- Each point corresponds to an observation

Data visualization

Scatterplot



Source: Horst A., et al. (2020) palmerpenguins: Palmer Archipelago (Antarctica) penguin data, <https://allisonhorst.github.io/palmerpenguins/>

Linear regression

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Linear Regression

With **regression** we will try to find a **consistent** and **systematic** relationship between two quantitative variables.

1. **Monotonic:** consistent direction of the relationship between the two variables: increasing or decreasing
2. **Non-monotonic:** value of dependent variable changes systematically with value of independent variable, but the direction is not consistent

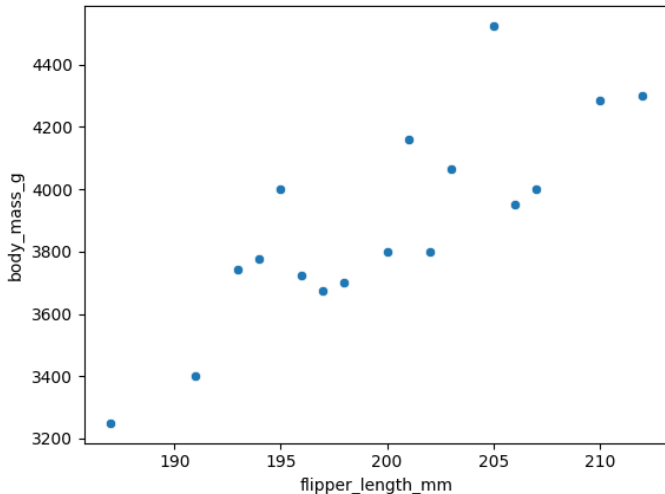
Linear Regression

A *linear* relationship between an independent and dependent variable.

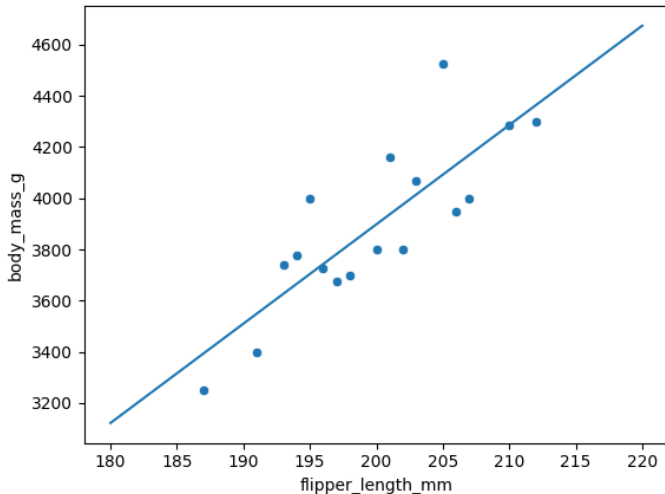
Characteristics:

- Presence: is there a relationship?
- Direction: increasing or decreasing?
- Strength of the relationship: strong, moderate, weak, nonexistent, ...

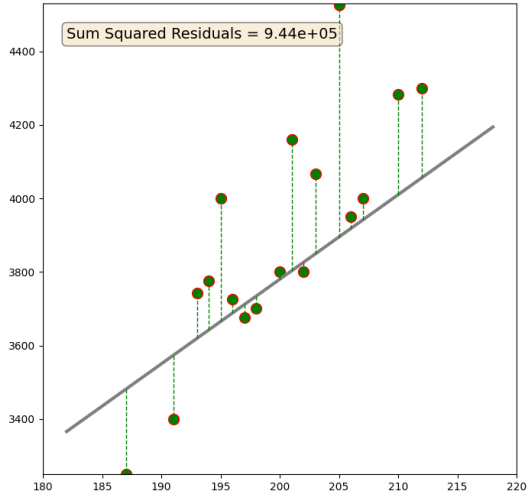
Linear Regression



Linear Regression

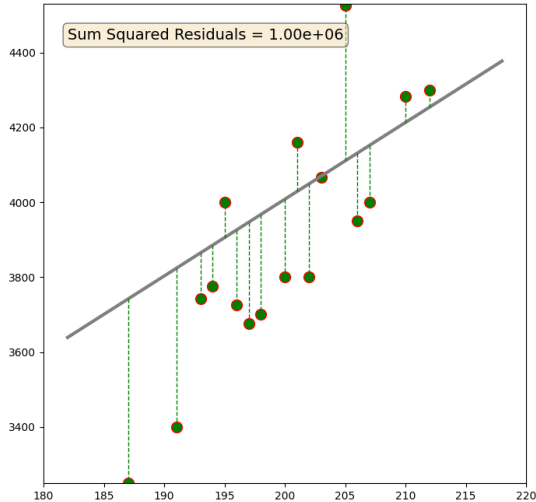


Method of least squares



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Method of least squares



Method of least squares

	x	y	$(x-\bar{x})$	$(y-\bar{y})$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$
0	187.0	3250.000000	-12.823529	-641.274510	8223.402537	164.442907
1	191.0	3400.000000	-8.823529	-491.274510	4334.775087	77.854671
2	193.0	3741.666667	-6.823529	-149.607843	1020.853518	46.560554
3	194.0	3775.000000	-5.823529	-116.274510	677.128028	33.913495
4	195.0	4000.000000	-4.823529	108.725490	-524.440600	23.266436
5	196.0	3725.000000	-3.823529	-166.274510	635.755479	14.619377
6	197.0	3675.000000	-2.823529	-216.274510	610.657439	7.972318
7	198.0	3700.000000	-1.823529	-101.274510	184.704604	3.325260

...

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Method of least squares

Equation

The regression line has the following equation:

$$\hat{y} = \beta_1 x + \beta_0$$

with:

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \approx \frac{29375.49}{756.47} \approx 38.83$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} \approx 3891.27 - 38.83 \times 199.82 \approx -3868.33$$

$$\hat{y} = 38.83x - 3868.33$$

Note: $\hat{y}$ indicates “an estimation for y ”

Covariance

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Covariance

Covariance

Covariance is a measure that indicates whether a relationship between two variables is increasing or decreasing.

$$\text{Cov}(X, Y) = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

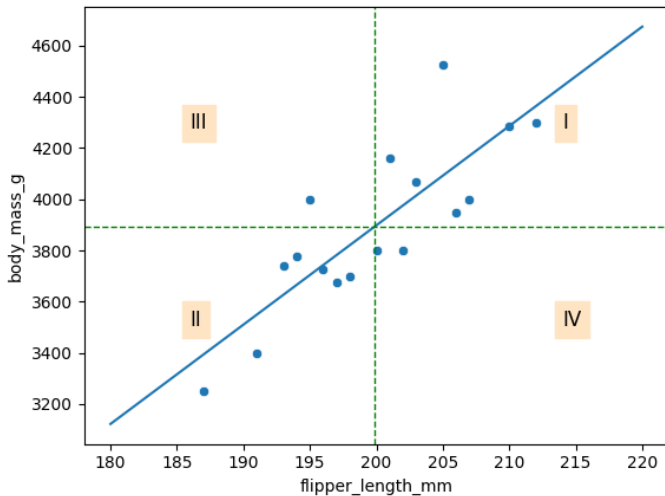
Cov > 0: increasing

Cov $\approx$ 0: no relationship

Cov < 0: decreasing

Note Covariance of population (denominator n)
vs. sample (denominator $n - 1$)

Covariance



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Covariance

$$\text{Cov}(X, Y) \approx \frac{29375.49}{17 - 1} \approx 1835.97$$

- $\text{Cov} > 0 \Rightarrow$ increasing relationship
- What if body mass was expressed in kg instead of g?

Covariance has limited use as a measure of the relationship between two variables.

Pearson's correlation coefficient

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Pearson correlation coefficient

Pearson's Correlation Coefficient

Pearson's product-moment correlation coefficient R is a measure for the strength of a linear correlation between x and y

$$R = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} \quad (1)$$

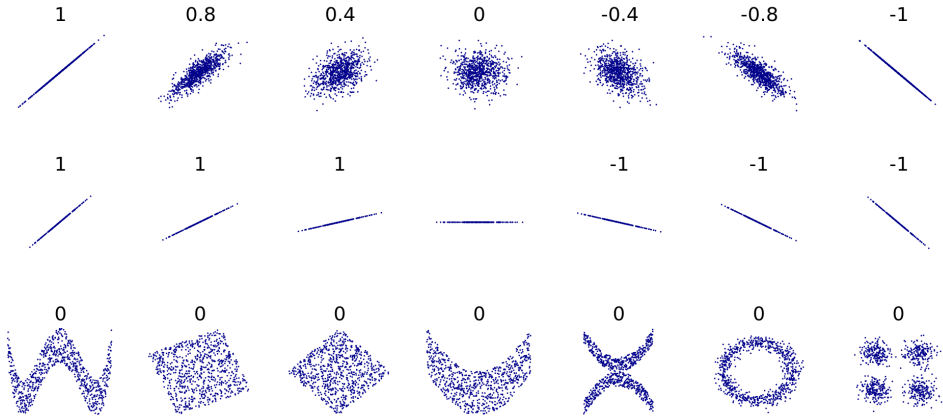
$$= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (2)$$

$$R \in [-1, +1]$$

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Correlation coefficient

Some datasets and their R-value



Source: Wikipedia https://en.wikipedia.org/wiki/Pearson_correlation_coefficient

Coefficient of determination

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Coefficient of determination

Coefficient of determination

The coefficient of determination R^2 explains the percentage of the variance of the observed values relative to the regression line.

R^2 : percentage variance observations explained by the regression line

$1 - R^2$: percentage variance observations *not* explained by regression

Interpretation of R and R^2 values

$ R $	R^2	Explained variance	Interpretation
< 0.3	< 0.1	< 10%	very weak
0.3 - 0.5	0.1 - 0.25	10 - 25%	weak
0.5 - 0.7	0.25 - 0.5	25 - 50%	moderate
0.7 - 0.85	0.5 - 0.75	50 - 75%	strong
0.85 - 0.95	0.75 - 0.9	75 - 90%	very strong
> 0.95	> 0.9	> 90%	exceptional(!)

Strength of relationship

Example chinstrap penguins

$$\begin{aligned} \text{Cov}(X, Y) &\approx \frac{29375.49}{17 - 1} \approx 1835.97 \\ R &= \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} \\ &\approx \frac{1835.968}{6.876 \times 322.935} \approx 0.827 \\ R^2 &\approx 0.827^2 \approx 0.684 \end{aligned}$$

Conclusion: There is a strong linear and increasing relationship between flipper length and body mass of male chinstrap penguins. 68.4% of the variance in body mass can be explained by the variance in flipper length.

Considerations

- The correlation coefficient only looks at the relationship between **two variables**. Interactions with other variables are not considered.
- The correlation coefficient explicitly does **not** assume a **causal** relationship.
- Pearson's correlation coefficient only expresses **linear** relationships.

Module 7. Time series analysis

Data Science & AI

Sabine De Vreese Lieven Smits Bert Van Vreckem
2023–2024

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Contents

- Time series and predictions

- Time series models

 - Mathematical model

 - General

 - Estimating the parameters

- Moving average

 - Simple moving average

 - Weighted moving average

 - Double exponential smoothing

 - Forecasting

 - Triple exponential smoothing

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Learning goals

- Concepts, time series models
- Moving average
- Exponential smoothing

Remark

You should not memorize the formulas in this module, but you should understand them!

Time series and predictions

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Time series and predictions

Time series

A time series is a sequence of observations of some variable over time.

- Monthly demand for milk
- Annual intake of students at HOGENT
- Price of a share or bond on the stock exchange (hourly, daily, ...)
- Number of HTTP requests per second for a website
- Evolution of disk usage on a backup server

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Time series and predictions

Many decisions in business operations depend on a forecast of some quantity

- General development of future plans (investments, capacity ...)
- Budget planning to avoid shortcomings (operating budget, marketing budget ...)
- Procurement planning (e.g. storage capacity)
- Support for financial objectives
- Avoid uncertainty

Time series and predictions

Time series are a **statistical** problem: observations vary with time

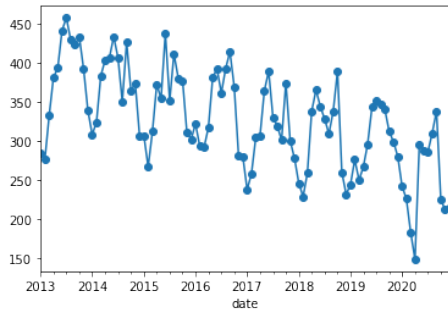
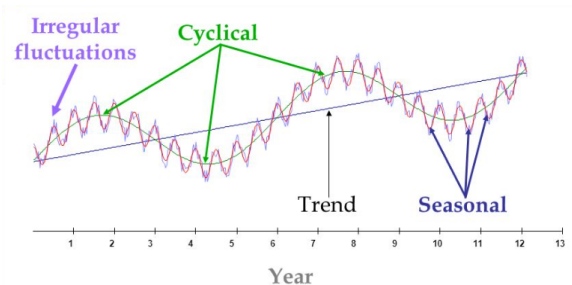


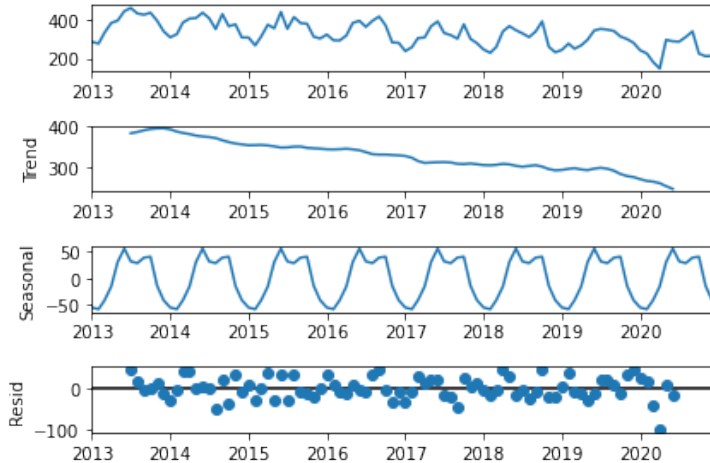
Figure: Number of heavily wounded in car accidents in Flanders.

Time series components

- Level
- Trend
- Seasonal fluctuations
- Cyclic patterns
- Random noise (residuals)



Time series decomposition



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Time series models

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Mathematical model time series

The simplest model:

$$X_t = b + \varepsilon_t \quad (1)$$

- X_t : estimate for time series, at time t
- b : the level (a constant), based on **observations** x_t
- ε_t : random **noise**. We assume that $\varepsilon_t \sim \text{Nor}(\mu = 0; \sigma)$

Mathematical model time series

We could also assume that there is a linear relationship:

$$X_t = b_0 + b_1 t + \varepsilon_t \quad (2)$$

with **level** b_0 and **trend** b_1 .

Equation 1 and 2 are special cases of the **polynomial** case:

$$X_t = b_0 + b_1 t + b_2 t^2 + \dots + b_n t^n + \varepsilon_t \quad (3)$$

General expression time series

$$X_t = f(b_0, b_1, b_2, \dots, b_n, t) + \varepsilon_t \quad (4)$$

We make these assumptions:

- We consider two components of variability:
 - the mean of the predictions changes with time
 - the variations to this mean vary randomly
- The residuals of the model ($X_t - x_t$) have a constant variance in time (**homoscedastic**)

Estimating the parameters

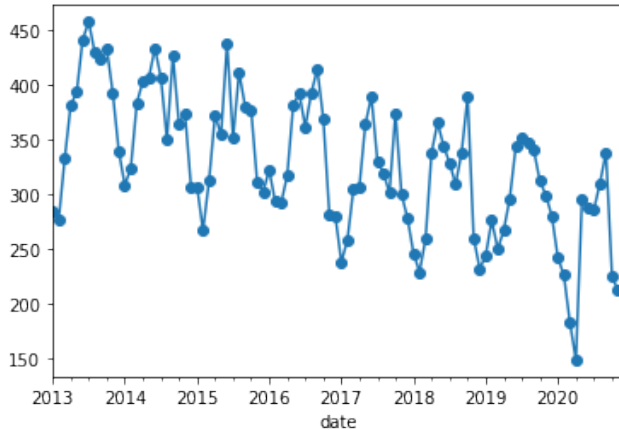
Make **predictions** based on the time series model:

1. select the most suitable model
2. estimation for parameters $b_i (i : 1, \dots, n)$ based on observations

The estimations $\hat{b}_i$ are selected so that they approximate the observed values as close as possible.

Example

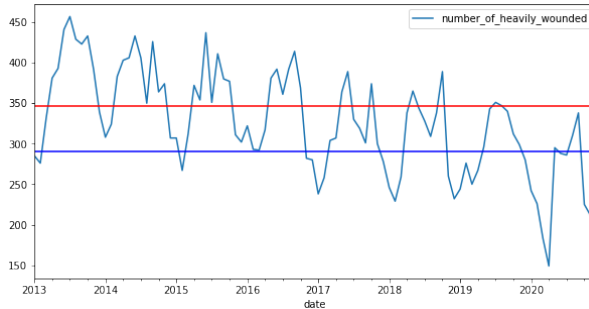
Number of heavily wounded in car accidents in Flanders



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Example: parameter estimation

- We select the constant model of Equation 1
- You choose which observations you use to determine $\hat{b}$, e.g.
 - First 70 observations: $\hat{b} = \frac{1}{70} \sum_{t=1}^{70} x_t = 346.4$
 - Last 50 observations: $\hat{b} = \frac{1}{50} \sum_{t=47}^{96} x_t = 290.68$

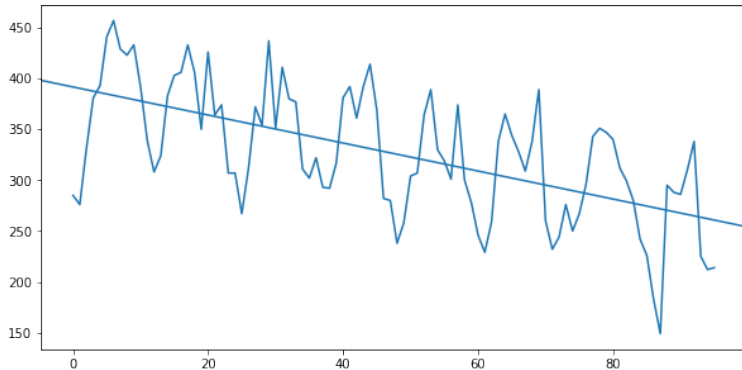


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Example: parameter estimation

If we want to model the observations with a linear function

$X_t = b_0 + b_1 t + \varepsilon_t$, then we can use linear regression!



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Moving average

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Moving average

Moving Average

The moving average is a series of averages (means) of the last m observations

- Notation: SMA
- Hide short-term fluctuations and show long-term trends
- Parameter m is the time window

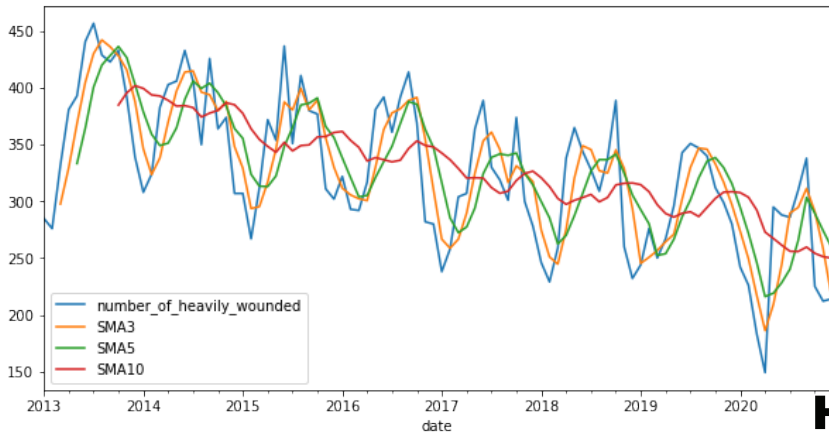
$$SMA(t) = \sum_{i=k}^t \frac{x_i}{m}$$

with $k = t - m + 1$.

(5)

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Example



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Example: “Golden cross”

Moving averages are used in the **technical analysis** of stock prices to discover trends:



Weighted moving average

- For *SMA*, the weights of the observations are equal
- For a weighted moving average (*WMA*), more recent observations gain relatively more weight
- A specific form of this is single exponential smoothing or the **exponential moving average** (EMA):

$$X_t = \alpha x_{t-1} + (1 - \alpha)X_{t-1} \quad (6)$$

with α the smoothing constant ($0 < \alpha < 1$), and $t \geq 3$

Exponential smoothing

Equation 6 is only valid from $t = 3$. Hence, we need to choose a suitable value for X_2 ourselves. There are several options:

- $X_2 = x_1$
- $X_2 = \frac{1}{m} \sum_{i=1}^m x_i$ (so the mean of the first m observations)
- make X_2 equal to a specific objective
- ...

Exponential Smoothing

Example of calculation ($\alpha = 0.1$)

t	1	2	3	4	...
x_t	8	10	11	13	...
X_t	—	8	8.2	8.68	...

$$X_1 = \text{undefined}$$

$$X_2 = x_1$$

$$X_3 = 0.1 \times 10 + 0.9 \times 8 = 8.2$$

$$X_4 = 0.1 \times 13 + 0.9 \times 8.2 = 8.68$$

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Why “*exponential*”?

$$\begin{aligned}X_t &= \alpha x_{t-1} + (1 - \alpha)X_{t-1} \\&= \alpha x_{t-1} + (1 - \alpha) [\alpha x_{t-2} + (1 - \alpha)X_{t-2}] \\&= \alpha x_{t-1} + \alpha(1 - \alpha)x_{t-2} + (1 - \alpha)^2 X_{t-2}\end{aligned}$$

or in general:

$$= \alpha \sum_{i=1}^{t-2} (1 - \alpha)^{i-1} x_{t-i} + (1 - \alpha)^{t-1} X_{t-1}, t \geq 2$$

In other words: older observations have an exponentially smaller weight.

Exponential smoothing

α	$(1 - \alpha)$	$(1 - \alpha)^2$	$(1 - \alpha)^3$	$(1 - \alpha)^4$
0.9	0.1	0.01	0.001	0.0001
0.5	0.5	0.25	0.125	0.062
0.1	0.9	0.81	0.729	0.6561

Table: Values for α and $(1 - \alpha)^n$

The speed at which the old observations are “forgotten” depends on the value of α . For a value of α close to 1, old observations are quickly forgotten, whereas for α close to 0, this goes less fast.

Example

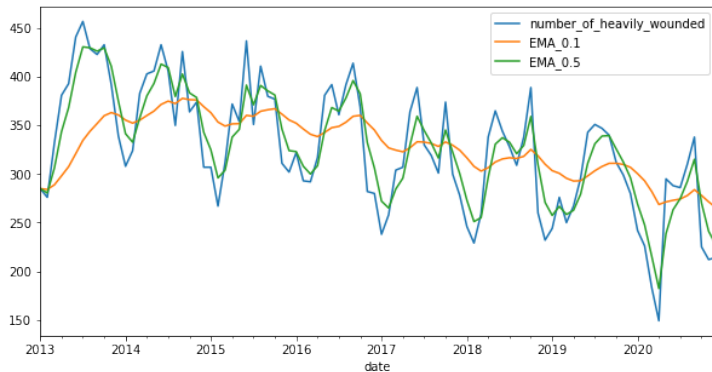


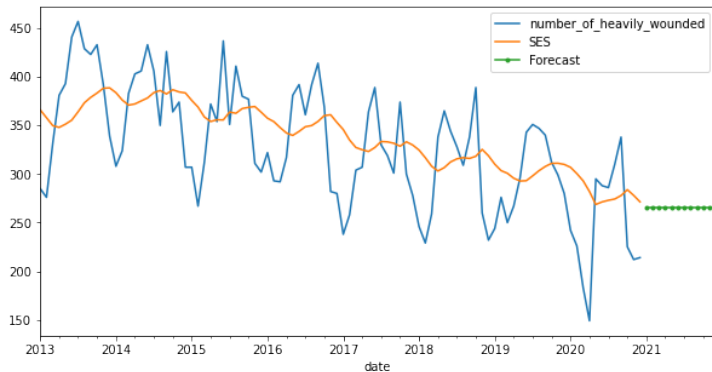
Figure: Single exponential smoothing with $\alpha = 0.1, 0.5$

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Forecasting

As a forecast for time $t + m$ (m time units in the “future”), we always take the last estimate of the level:

$$F(t + m) = X_t$$



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Double exponential smoothing

Basic exponential smoothing does not work well if there is a trend in the data

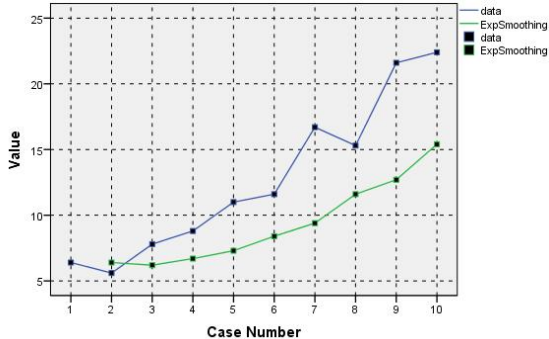


Figure: Exponential smoothing with a trend: the errors keep getting bigger

Double exponential smoothing

We add an additional term to model the trend. We use b_t for the estimation of the trend at time $t > 1$:

$$X_t = \alpha x_t + (1 - \alpha)(X_{t-1} + b_{t-1})$$

$$b_t = \beta(X_t - X_{t-1}) + (1 - \beta)b_{t-1}$$

with $0 < \alpha < 1$ and $0 < \beta < 1$

- b_t is an estimate for the slope of the trend line
- Added to the first equation to ensure that the trend is followed
- $X_t - X_{t-1}$ is positive or negative, this corresponds to an increasing/decreasing trend

Double exponential smoothing

Again, there are different options for selecting the initial values:

$$X_1 = x_1$$

$$b_1 = x_2 - x_1$$

$$b_1 = \frac{1}{3} [(x_2 - x_1) + (x_1 - x_2) + (x_4 - x_3)]$$

$$b_1 = \frac{x_n - x_1}{n - 1}$$

Predicting (forecasting)

To make a prediction (forecast) $F(t + 1)$ for time $t + 1$ we use:

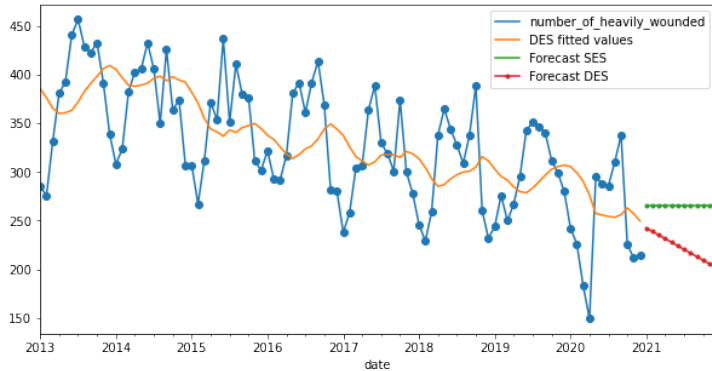
$$F(t + 1) = X_t + b_t$$

or in general for time $t + m$:

$$F(t + m) = X_t + mb_t$$

Example

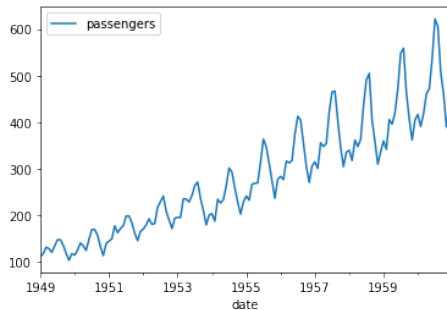
Comparison of Single/Double Exponential Smoothing forecasts



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Triple exponential smoothing

Some time series have recurring (seasonal) patterns, e.g.



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Triple exponential smoothing

Or Holt-Winter's Method

Notation:

- L : length of the seasonal cycle (number of time units)
- c_t : term that models the seasonal variations
- γ (gamma): smoothing factor for the seasonal variation

$$X_t = \alpha \frac{X_t}{c_{t-L}} + (1 - \alpha)(X_{t-1} + b_{t-1})$$

Smoothing

$$b_t = \beta(X_t - X_{t-1}) + (1 - \beta)b_{t-1}$$

Trend smoothing

$$c_t = \gamma \frac{X_t}{X_t} + (1 - \gamma)c_{t-L}$$

Seasonal smoothing

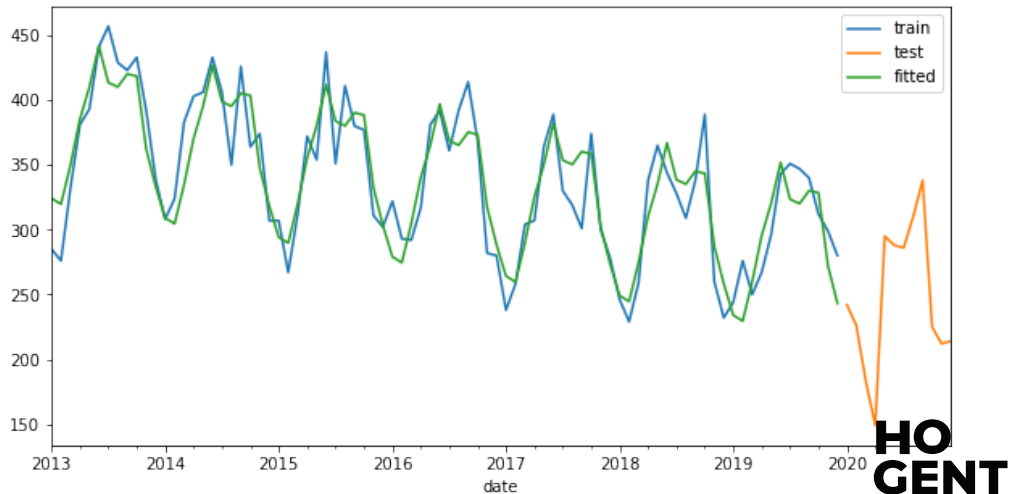
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Triple exponential smoothing

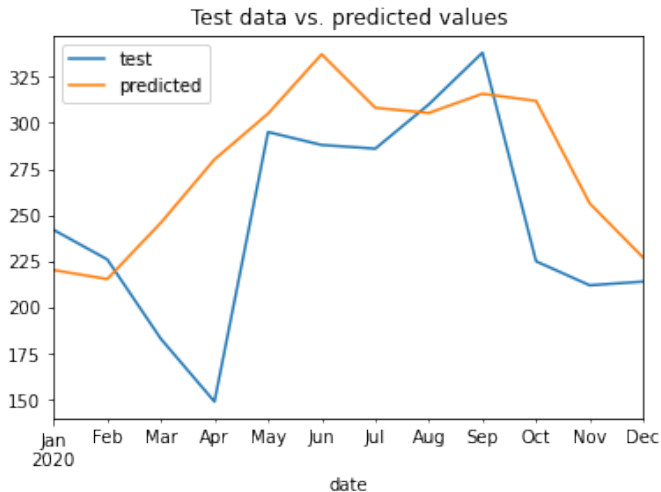
Prediction at time $t + m$:

$$F_{t+m} = (X_t + mb_t)c_{t-L+m}$$

Example



Quality of a time series model



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Quality of a time series model

Compare forecast results with actual observations, when they become available:

- Mean absolute error: $MAE = \frac{1}{m} \sum_{i=t+1}^{t+m} |x_i - F_i|$
- Mean squared error $MSE = \frac{1}{m} \sum_{i=t+1}^{t+m} (x_i - F_i)^2$

If square root of MSE is well below standard deviation over all observations, you have a good model!